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NEP 2020 FOLLOWS NCF-SE

A Textbook of
Computer Science with Lab Activities

Mastering CODING

AI AND ROBOTICS

Windows 10 & MS Office 2016

PREPARATORY STAGE 4

Scratch

ScratchJr

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Mobile Application | Video Learning | AI- Powered Worksheets | Test Paper Generator

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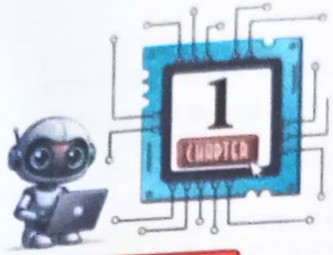


History of Computers

LET'S RECALL

Write the names of different types of computer shown below :

IN THIS CHAPTER

- The Beginning of Computers
- The Slide Rule
- Difference Engine
- The Abacus
- Pascaline
- Analytical Engine
- Calculating Rods
- Leibniz Calculating Machine

As we know, computer is an electronic machine that can think and do many tasks for us. It can help us write stories, play games, watch videos, and even solve math problems. We tell the computer what to do, and it does it quickly.

Computers are very important in our daily lives. We use them in schools to learn new things, at home to play games or watch cartoons, and even in shops to buy things. Computers help doctors in hospitals, scientists in labs, and people in offices do their work better and faster. Without computers, many of the things we enjoy and rely on today would be much harder to do.

BEGINNING OF COMPUTERS

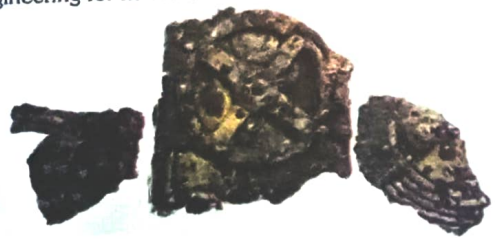
In ancient times, people used different methods to count and calculate. If we look back 40 years, the computers in use were very different from the ones we see today.

Long ago, ancient people counted using their fingers and toes. They also used small stones or made marks on sticks to keep track of numbers. Sometimes, they traded things like shells or beads to show amounts. This was a clever way to do maths without modern tools.

Now, it's time to explore the history of computers. Let's start!

INFO Byte

The Antikythera mechanism is an ancient device, often called the first analogue computer. Found in a shipwreck near Antikythera (island in Greece) in 1901, it dates back to around 150-100 BCE. It was used to predict the positions of stars and eclipses and could even calculate the living of the Olympic Games. Its advanced gears show incredible engineering for its time.



The Abacus

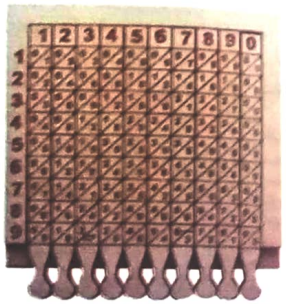
The abacus is a simple counting tool made of a wooden frame with beads on rods. People used it to add and subtract numbers by moving the beads back and forth. It was invented thousands of years ago by the Chinese, long before there were any electronic computers. The abacus was the first step in the history of computers because it helped people solve maths problems more easily.

Abacus

Even though the abacus is a basic tool compared to today's computers, it was an important invention. It showed that devices could help us think and work faster, which led to the development of the more advanced computers we use now.

Calculating Rods (Napier's Bones)

Calculating rods, also known as Napier's Bones, were tools used a long time ago to help people do Maths. Invented by John Napier in the early 1600s, these rods were thin sticks marked with numbers. By arranging the rods in different ways, people could add, subtract, multiply, and divide numbers. These rods made it easier to solve maths problems before there were calculators



Calculating Rods

or computers. They were like an early calculator that people could carry with them.

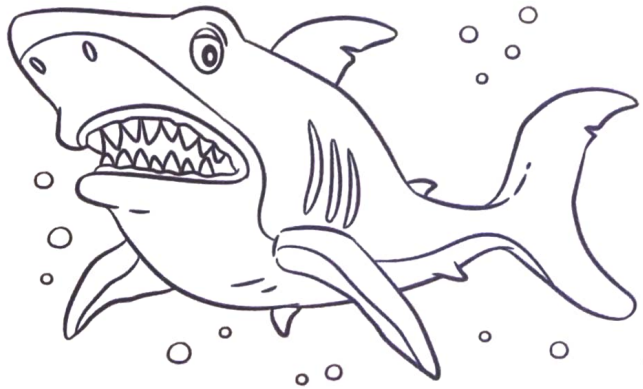
The Slide Rule

The slide rule is a tool that looks like a ruler, but is used for more than just measuring. It was invented by William Oughtred and consists of two parts that slide past each other, featuring special markings. By moving the parts and lining up the markings, people could do calculations like multiplication and division. Scientists and engineers used slide rulers to perform complex math quickly before electronic calculators became available.

Slide Rule

Activity Corner

This 400-years-old shark is the oldest living creature on the Earth. Colour it.



Pascaline

The Pascaline was one of the first machines that could do maths. It was invented by a French mathematician named Blaise Pascal in 1642. The Pascaline featured a series of wheels or dials, and each wheel represented a digit (like in a number). When you turned the wheels, it could add

Pascaline Machine

or subtract numbers. It was a very important invention because it was one of the earliest steps toward creating the computers we use today. Although it was simple compared to modern computers, the Pascaline made maths easier and faster for people at that time.

Leibniz Calculating Machine (Stepped Reckoner)

The Leibniz Calculating Machine, also known as the Stepped Reckoner, was invented by a German mathematician named Gottfried Wilhelm Leibniz in 1671. This machine was more advanced than the Pascaline, as it could do more than just add and subtract—it could also multiply and divide.

Leibniz Calculating Machine

The machine had a series of gears and levers that could be turned to perform different calculations. By moving these parts, the Leibniz machine could solve more complex maths problems than earlier tools. It was an important step in the development of mechanical calculators, which eventually led to the creation of modern computers.

Difference Engine

The Difference Engine, designed by Charles Babbage in the 1820s, was an early mechanical computer. It used gears and levers to do complex calculations automatically, which was much faster and more accurate than manual calculations. Although it was never fully built at the time, the Difference Engine was an important step towards the development of modern computers.

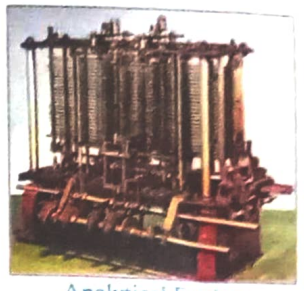
Difference Engine

Let's Get Better

When the first computer was so big, then how the modern computers are so small in size?

Analytical Engine

The Analytical Engine, also designed by Charles Babbage in the 1830s, was a more advanced machine than the Difference Engine. It was one of the first designs for a general-purpose computer. The Analytical Engine could perform any type of calculation and was programmable, meaning it could follow a series of instructions to solve different problems. Though it was never completed, it had many features that are used in modern computers, like a memory and a way to process instructions. It was a big step towards the computers we use today. That is why Charles Babbage is often called the 'Father of Modern Computers'.



Analytical Engine

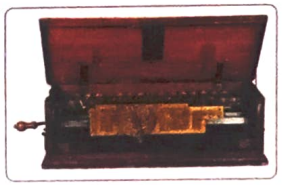
I. Fill in the blanks :

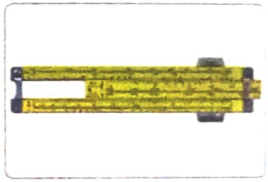
1. ..... is the world's oldest computing device.
2. ..... was invented by William Oughtred.
3. The ..... is known as the first mechanical and automatic calculator.
4. Difference engine was developed by ..... .
5. Concept of modern computers are based on the design of ..... .

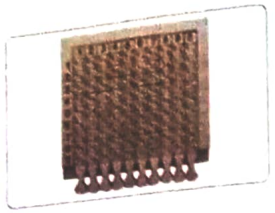
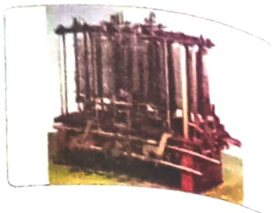
II. Answer the following questions :

1. Write a short note on Slide Rule.
2. Name the machine invented by John Napier and write a few words about it.
3. What is the Step Reckoner and who invented it?
4. Name two machines which were developed by Charles Babbage?

III. Identify the following images and write their names :





IV. Find pictures of the machines that were invented prior to the present day personal computers. Prepare a collage using the same.

Lab Activity

- Find out the details about the first computer made by HCL in India.
- Find out about the history of computer games, you play on the computer.
- List different personalities involved in the development of computer and related technologies.

CHAPTER 2

GUI Operating System-Windows 10

Let's Recall

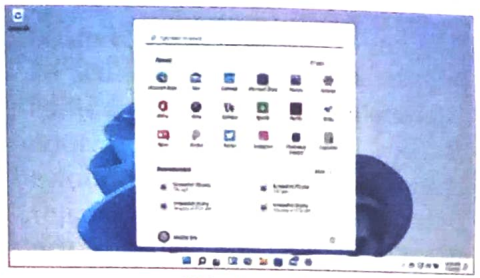
Write the names of three operating systems.

1. ..... 2. ..... 3. .....

In this Chapter

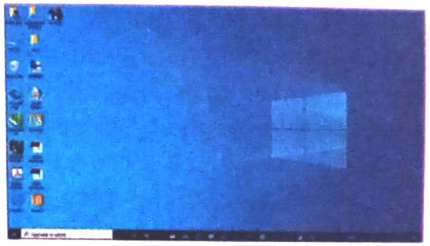
- Operating System
- Windows 10 as an Operating System
- Customising Desktop
- Functions of an Operating System
- Features of Windows 10
- Displaying multiple application windows on the Desktop

When you turn on your computer, you'll notice that it doesn't start up right away. You have to wait a few seconds before it's ready to use. During this time, a group of programs called the Operating System is getting the computer ready for you. Once it's done, the main screen that appears is called the Desktop.



Windows 11

Windows 8



Windows 10

INFO BYTE

An operating system is a set of programs that helps a computer run smoothly. It manages all the activities of the computer and makes it easier for you to use.

Functions of an Operating System

The main functions of an operating system include :

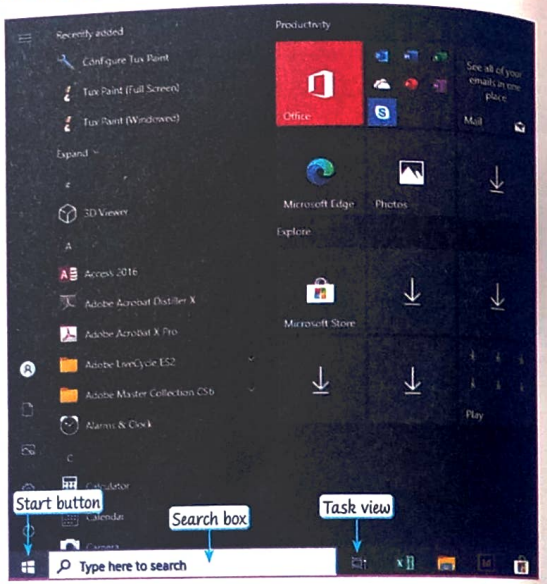
- It manages communication between hardware and software.
- It organises and assigns tasks to different software programs.
- It understands and follows the commands you provide.
- It detects and reports any errors in the computer system.

INFO BYTE

In a 'Graphical User Interface' (GUI) operating system, you can do tasks by simply clicking on available options. These tasks are usually app-based, so you don't need to remember commands. Windows 8, Windows 10, and Windows 11 are some of the most popular GUI-based operating systems.

WINDOWS 10 AS AN OPERATING SYSTEM

Windows 10 is an advanced operating system developed by Microsoft in 2015. It combines the best features of the previous versions including, Windows 7 and Windows 8. Some major changes include the return of the Start button in the lower left-hand corner, Live Tiles inside the Start menu, and a search box next to the Start button. It also introduces a new browser called Microsoft Edge and Cortana, a chat-based assistant. The Task View feature lets you see all your open windows on one screen.



You can also personalize your desktop by organising apps. The platform shows real-time information using tiles. For example, the tile for the weather app updates regularly to show the current temperature in your city.

INFO BYTE

Like Apple's Siri you can ask anything from Microsoft's 'Digital Assistant' known as Cortana.

Features of Windows 10

Here are some features of Windows 10 as an operating system :

- It offers a user-friendly environment.
- The Start menu is well-organised.
- It supports multimedia.
- It includes a new browser called Microsoft Edge.
- It also supports MS-DOS (Command Prompt) for users who want to work in DOS mode.

INFO BYTE

The Start menu in Windows 10 features Live Tiles, which can show dynamic content such as news updates, weather forecasts, and calendar events.

INFO BYTE

Windows 10 was officially released on July 29, 2015, as a free upgrade for users of Windows 7 and Windows 8.1.

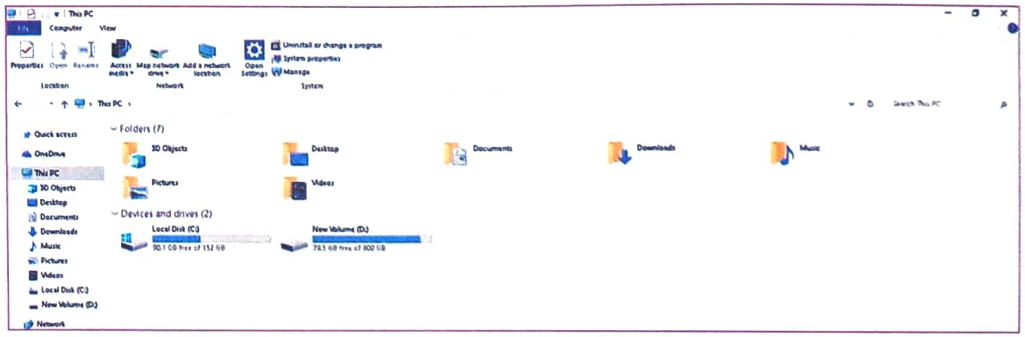
This PC (System Icon)

This PC is one of the most important icons in the Windows 10 operating system, usually found on the left side of the desktop. When you double-click on it, it shows details about the drives in your system.

This PC

This PC window displays :

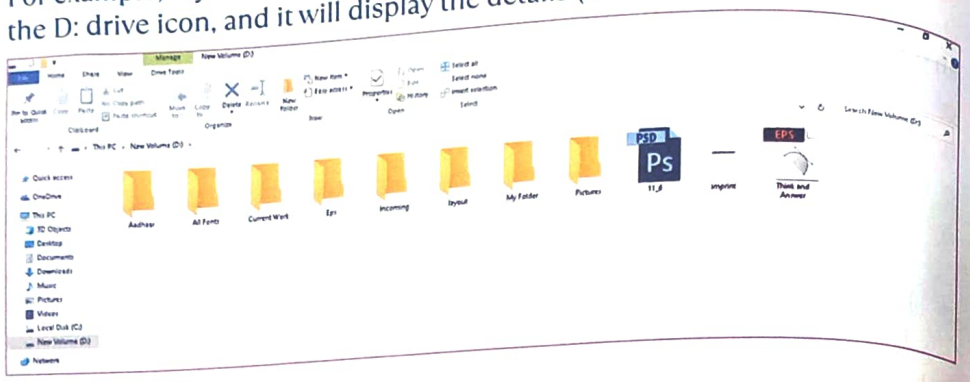
- The different drives in your system, such as the C: drive, D: drive, E: drive, and F: drive.



- Information on each drive's storage capacity, how much space is used, and how much is available.
- A DVD-RW drive, which enables you to read and write data on a CD or DVD.

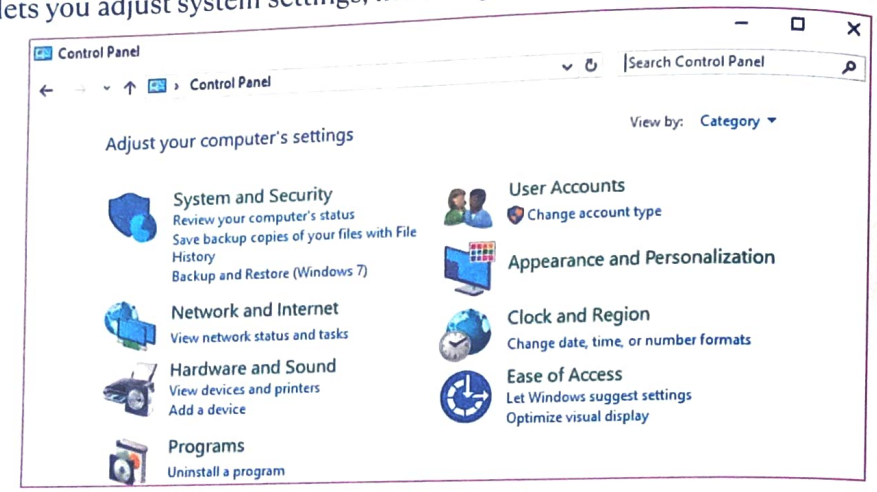
To view the contents of a specific drive, simply double-click the drive icon. This will show you all the files and folders stored in it.

For example, if you want to see the contents of the D: drive, just double-click the D: drive icon, and it will display the details (as shown below).



Control Panel

The Control Panel is the central hub for managing your computer's settings. It lets you adjust system settings, including adding or removing hardware and software. Hardware settings in the Control Panel include options for Display, Keyboard, Mouse, etc., while software settings cover things like Date and Time, Fonts, etc.



To open the Control Panel, double-click the Control Panel icon on your desktop. The Control Panel window will open, displaying all the available settings. You can choose any of these options to modify system settings or uninstall programs as needed.

Activity Corner

When you have plugged a pen drive into your laptop/desktop then you noticed that the picture is appearing on the screen but no sound is being heard. Now, check the laptop and find out whether the sound is mute or not. If yes, then unmute the speakers and enjoy the music.

CUSTOMIZING DESKTOP

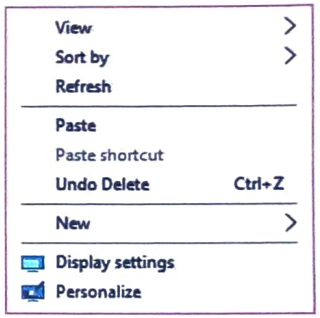
You can change the appearance of your Desktop from the default settings. You can customize the Desktop to suit your preferences, especially by changing the desktop background.

Background

To change the desktop background, follow these steps :

Step 1. Right-click a blank area on the desktop.

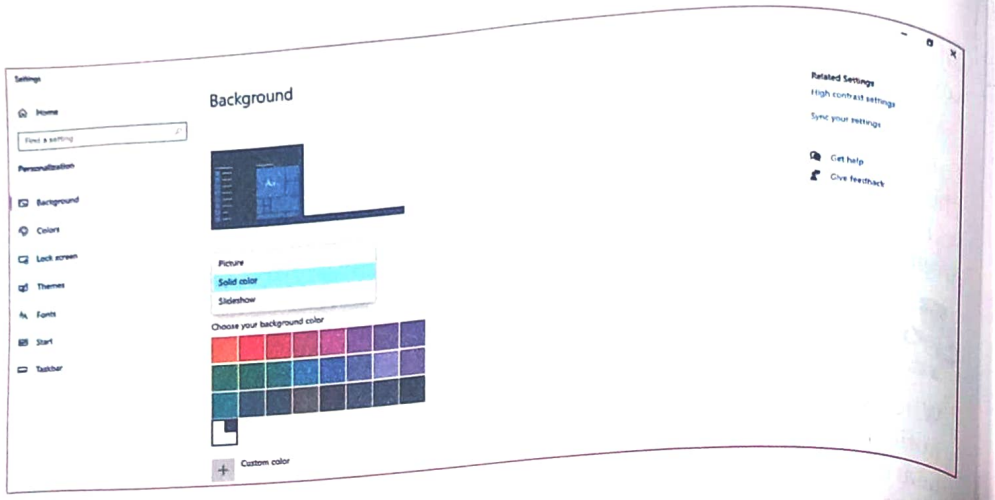
Step 2. Select Personalize from the menu that appears.



The Settings window will open, showing two sections. By default, the Background option on the left side is selected, and its details will appear on the right side.

Click the drop-down arrow (v) under the Background list, and you'll see the following options :

- Picture
- Solid Color
- Slideshow



Picture

Since the Background option is set to Picture, you'll see a list of pictures under this category. Click any picture from the list to select it, then close the window. The chosen picture will become your desktop background.

Solid Color

Click the drop-down arrow (v) under Background and choose Solid Color.

You'll see a list of available colours. Select the colour you like and then close the window. The chosen colour will become your desktop background.

Slideshow

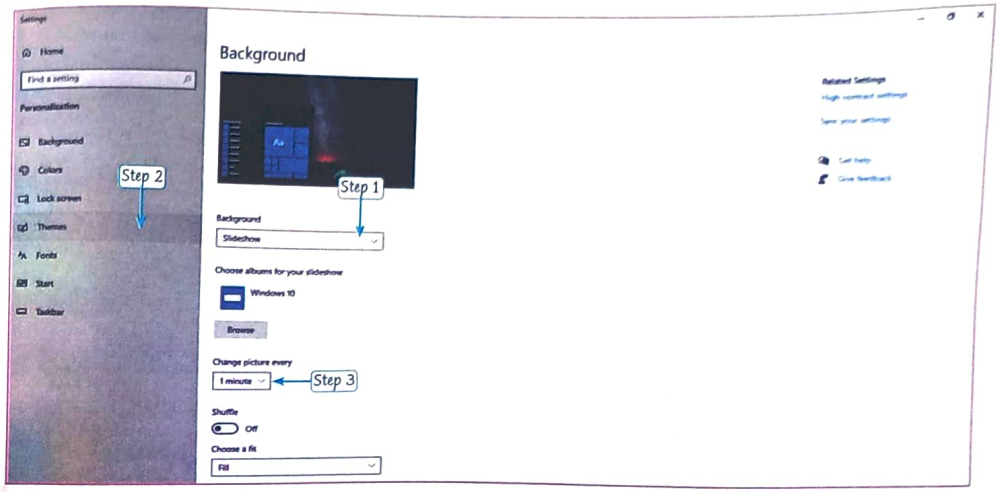
Before choosing Slideshow, firstly select the Themes. To do so, perform the following steps :

Step 1. Click Themes in the left pane to see all the available themes.

Step 2. Click the drop-down arrow (v) under Background and select Slideshow. Then, click the button to choose the folder with the images you want for the slideshow. The slideshow will show all the pictures in that folder.

Step 3. Use the Change picture every dropdown menu to set how long each picture will stay on the desktop (e.g., 1 minute). This means each picture will change every minute while the computer is idle.

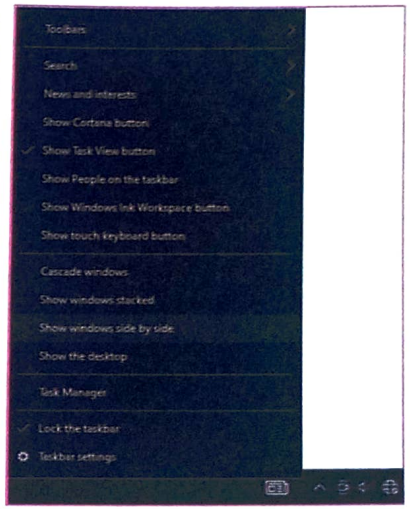
Step 4. Close the window.



You'll see the desktop background change at regular intervals as the slideshow runs.

DISPLAYING MULTIPLE APPLICATION WINDOWS ON THE DESKTOP

In Windows 10, you can display multiple application windows side by side, allowing you to work on them simultaneously. Here's how to do it :

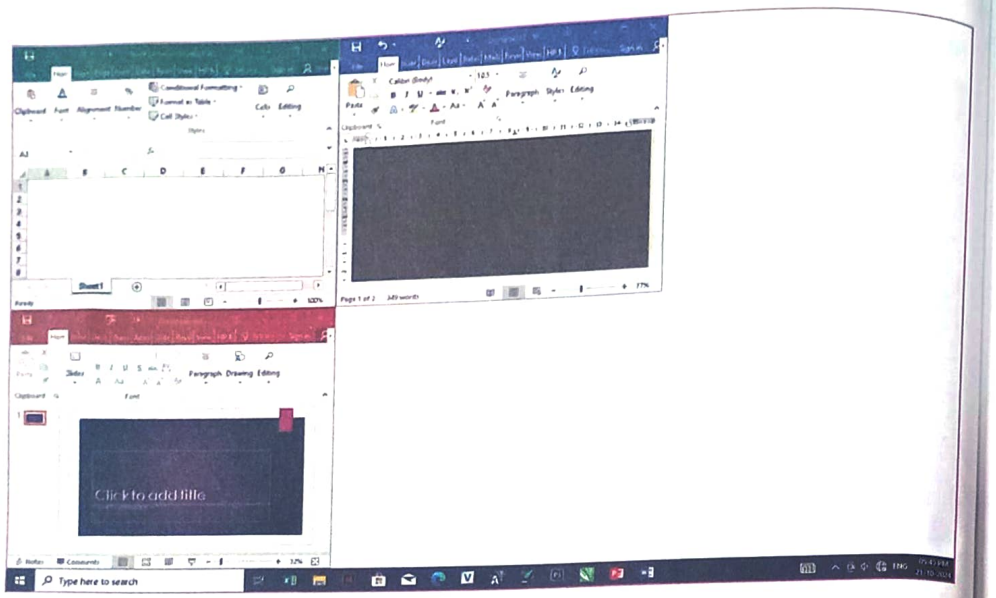


Step 1. Open two or more applications, such as MS Word 2016, MS Excel 2016, and PowerPoint 2016.

Step 2. Right-click on an empty area of the taskbar.

Step 3. Select Show windows side by side.

All your open application windows will now be displayed next to each other.



Shortcut

A shortcut is an icon that provides a quick and easy way to open frequently used applications or folders. You can recognise a shortcut icon by the small arrow in the bottom left corner. Double-clicking the shortcut icon will open the application or file you need.

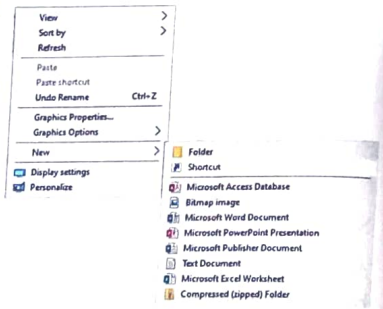
To create a shortcut on the desktop, follow these steps :

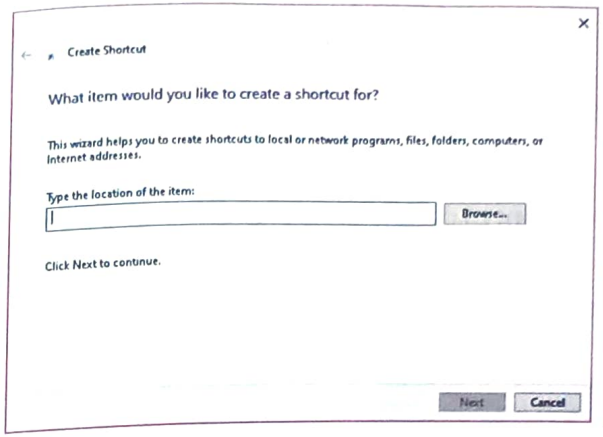
Step 1. Right-click on a blank area of the desktop.

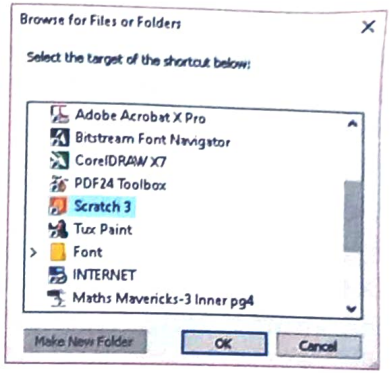
Step 2. Select New option from the menu that appears.

Step 3. Click on Shortcut from the submenu. A Create Shortcut dialog box will open.

Step 4. Click Browse to find the application or file you want to create a shortcut for. A new dialog box will open.



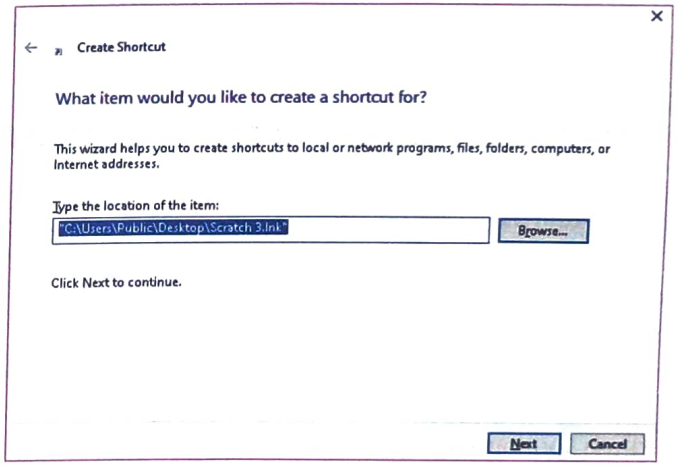




Step 5. Choose the application or file you want and click OK.

Step 6. The location of the selected item will show up in the location box. Click Next to continue.

Step 7. Type a name for the shortcut if you want to change the default name.

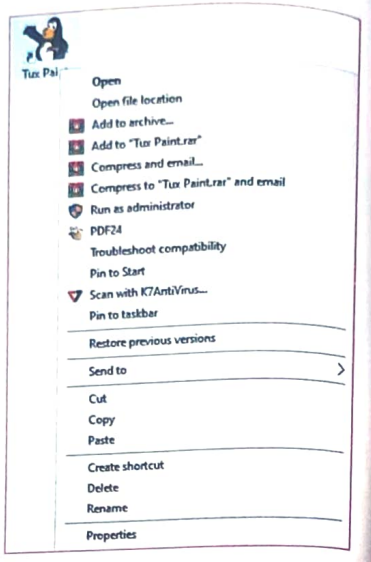


Step 8. Click Finish.

The new shortcut will now appear on your desktop with the name you specified.

Deleting Shortcut Icons from the Desktop

Shortcut icons on the desktop help you quickly access files, documents, or programs. If you find that some icons are no longer needed, you can delete them. Here's how :



Step 1. Right-click on the icon you want to delete.

Step 2. Select Delete from the menu.

The shortcut icon will be removed from your desktop.

SHORTCUT SECTION The shortcut key for delete a folder, a file or an icon is Delete or Del. If you want to permanently delete an item without sending it to Recycle Bin, use Shit + Delete.

Taskbar

The taskbar is a part of the Graphical User Interface (GUI) that appears as a long bar at the bottom of the screen. It has buttons and shortcut icons for apps that are pinned to it, allowing you to quickly access and perform tasks.

You'll also see the Start button on the far left and the system tray on the far right.

Typically, the taskbar holds shortcuts for frequently used programs, as well as buttons for any files, folders, or programs you are currently using.

Let's Get Better The Taskbar can help you switch between open programs without closing them. Just click on the icons at the bottom of your screen.

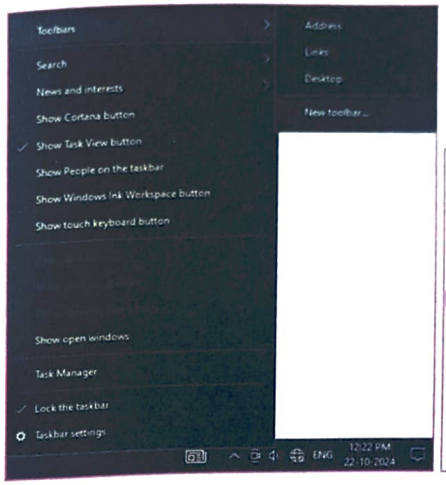
Quick Launch Toolbar

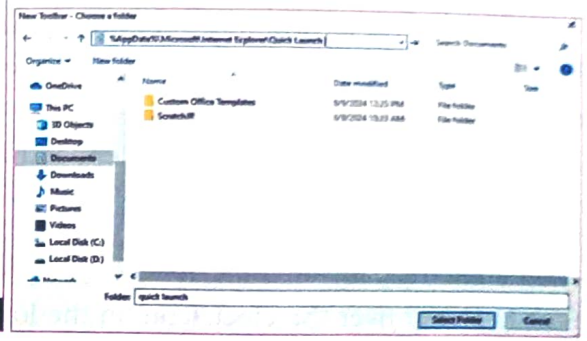
The Quick Launch toolbar was a feature in Windows XP that made it easy to access frequently used applications from the taskbar. Although it was removed in Windows 7, you can still add it to the taskbar if needed. Here's how :

Step 1. Right-click on an empty area of the taskbar.

Step 2. Choose Toolbars and then New toolbar... from the menu.

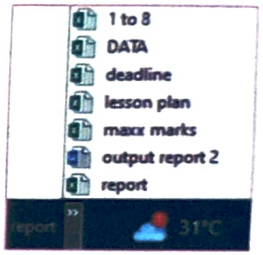
Step 3. Type % AppData% \ Microsoft \Internet Explorer\Quick Launch into the address bar of the New Toolbar window.





Step 4. Click Select Folder.

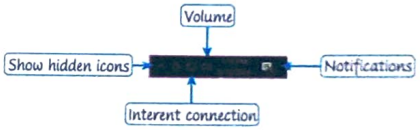
The Quick Launch toolbar will now appear on the taskbar. By default, it includes features like Show Desktop and Switch between windows icons.



System Tray

The notification area, also known as the system tray, is located on the far right side of the taskbar. It shows important icons by default, such as volume control, network status, and battery or charging information (if your device is plugged in).

It also has some hidden icons that you can see by clicking the up arrow (^) button. When you install new programs, like an antivirus, an icon will automatically appear in this area to notify you.



Customising Taskbar

By default, the taskbar is at the bottom of the desktop, but you can move it to a

different position, add or remove icons, or even hide it. This process is known as customising the taskbar.

Adding Icons on the Taskbar

To add an icon to the taskbar, follow these steps:

Step 1. Find the icon you want to add to the taskbar.

Step 2. Right-click on the icon.

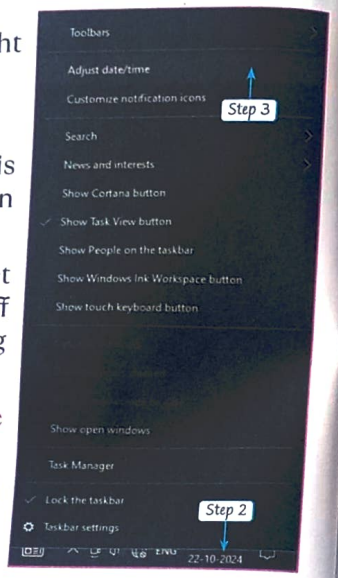
Step 3. Choose Pin to taskbar from the menu.

The selected icon will now appear on the taskbar.

Date and Time on Computer System

A computer system has a built-in calendar that shows the date and time on the far right side of the taskbar. When you hover over the clock icon, it shows the full day and date.

To change the date and time, follow these steps :



Step 1. Hover over the clock icon on the lower right side of the taskbar.

Step 2. Right-click on the clock icon.

Step 3. Select Adjust date/time from the menu. This will open a Settings window where you can set the date and time.

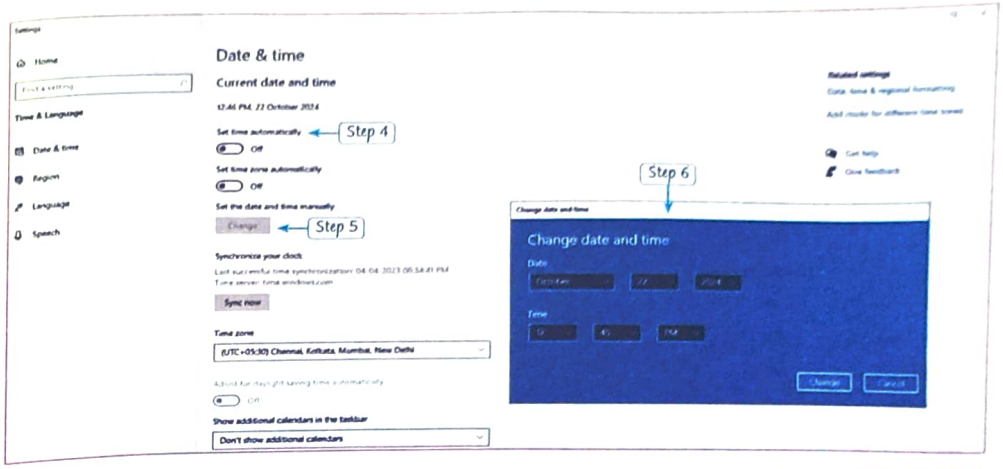
Step 4. By default, the date and time are set automatically. To set them manually, turn off the Set time automatically option by toggling it off (it will turn blue when disabled).

Step 5. Click Change under the Change date and time option.

Step 6. In the pop-up box, enter the new date and time you want.

Step 7. Click Change to update the date and time.

Step 8. Finally, turn the Set time automatically option back on to have Windows update the date and time automatically.

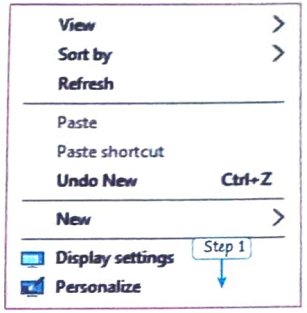


Note : The system clock uses a battery to keep the date and time accurate even when the computer is turned off.

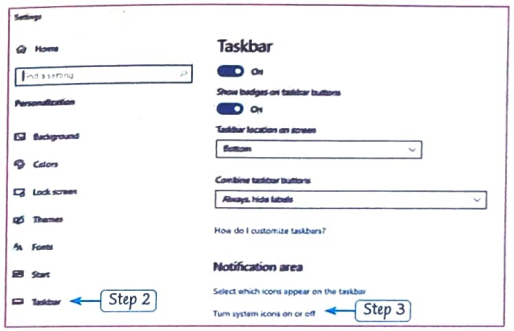
Hiding/Unhiding System Clock

A computer system always shows certain icons like the clock, volume, network, and power on the lower right side of the taskbar. These icons help you keep track of time and system status. You can choose to show or hide these icons. To hide or unhide the system clock, follow these steps :

Step 1. Right-click on a blank area of the desktop and select Personalize from the menu.

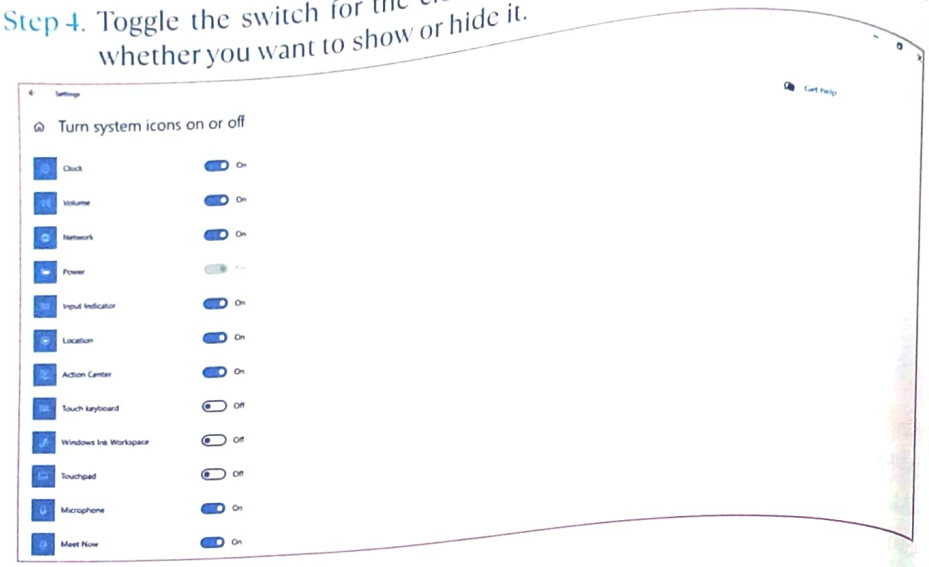


Step 2. Click on Taskbar in the left pane.



Step 3. Select Turn system icons on or off.

Step 4. Toggle the switch for the clock to turn it ON or OFF, depending on whether you want to show or hide it.



Step 5. Click OK to save your changes.

You can use the same steps to hide or unhide other system icons.

Volume of the Speaker

The Speakers icon is on the right side of the taskbar. To adjust the volume for playing audio or video, follow these steps :

Step 1. Click on the Speakers icon. A small window will pop up showing the current volume level.

Step 2. To change the volume, move the slider left to lower the volume or right to increase it.

If you want to adjust the volume for both the speakers and system sounds separately, right-click the Speakers icon and select Open Volume Mixer. This will open a window where you can control the volume for both the speakers and system sounds individually.

I. Tick (✓) the correct option :

1. Which of the following components of the desktop is illustrated alongside?

   (a) Dustbin ☐
   (b) Basket ☐
   (c) Recycle Bin ☐
   (d) Bucket ☐

2. The first screen of the Windows is called ..... .
   (a) Tabletop ☐ (b) Desktop ☐
   (c) Palmtop ☐ (d) Laptop ☐

3. Which of the following is not an operating system?
   (a) Windows 10 ☐ (b) Windows 8 ☐
   (c) Windows 11 ☐ (d) Windows 9 ☐

4. Which of the following icons will you use to uninstall a program from a computer system?
   (a) Library ☐ (b) Recycle Bin ☐
   (c) Control Panel ☐ (d) This PC ☐

5. The long horizontal bar available at the bottom of the screen is called ..... .
   (a) Taskbar ☐ (b) Sidebar ☐
   (c) Bottom bar ☐ (d) Title bar ☐

6. Which of the following icons will display all the drives available in a computer system?
   (a) Control Panel ☐ (b) This PC ☐
   (c) A folder ☐ (d) A shortcut ☐

II. Write 'T' for true and 'F' for false :

1. The deleted files, folders or icons are stored in documents. ☐
2. You can see the time displayed on the left hand side of the taskbar. ☐

3. Window is a rectangular area made available for the user on the screen.
4. The control panel is used to change settings and add or remove programs.
5. The sound can be made on and off by clicking on Speakers icon.

III. Fill in the blanks :

1. Windows 10 is also known as ..... Interface operating system.
2. ..... option allows you to set background colour of the desktop.
3. Quick Launch bar is part of ..... .
4. The ..... tile provides access to frequently updated information such as news, weather and calendar events.
5. The Speakers icon is located on the ..... side of taskbar.

IV. Name the following :

1. Two GUI based operating system :

   (a) ..... (b) .....
2. Two tasks to customise desktop :

   (a) ..... (b) .....
3. Two application programs (apps) pinned to taskbar :

   (a) ..... (b) .....

V. Write short notes on each of the following :

1. Operating system
2. This PC (System Icon)
3. Taskbar
4. Quick Launch bar
5. Shortcut to an icon
6. System Tray

VI. Write down the steps to do the following Windows operations :

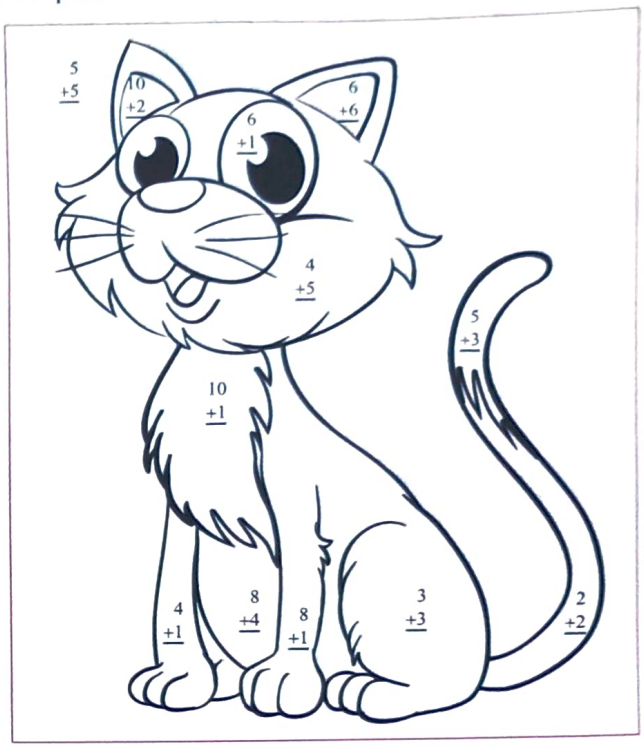
1. Displaying multiple applications on the desktop.
2. Adding icons on the Taskbar.
3. Creating a shortcut of MS Paint on the desktop.

4. Setting date and time of system clock.
5. Adjusting the volumes of speakers.

Smart Thinking

Colour by code :

4 = Blue
5 = Light Green
6 = Brown
7 = White
8 = Yellow
9 = Pink
10 = Gray
11 = Green
12 = Red

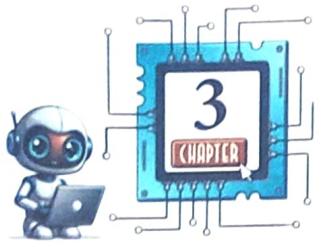


Lab Activity

In the computer lab, Windows 8 as well as Windows 10 operating systems are loaded. Now, make a project by comparing the features of Windows 10 with Windows 8 operating system. Finally, submit your project for the evaluation.

FOR THE TEACHER

- Explain the various components of the desktop, the start menu, how to work with tiles.
- Discuss the various icons and the system icons on the desktop.
- Demonstrate the students about changing the window colour and background.



Advanced Features of Word 2016

Let's Recall

Which of the following buttons is used to align the text to the left?

In this Chapter

- Text Formatting Tools
- Page Formatting Tools
- Text Editing Tools

Word 2016 includes advanced features that help you create professional-looking documents. It offers various tools that make it easy to format the text in your documents.

TEXT FORMATTING TOOLS

You've already learned about some formatting commands in your previous class, such as :

- Font : This is how the letters appear on the screen.
- Bold : This makes the text darker.
- Italics : This makes the text slanted.
- Underline : This adds a line under the text.
- Alignment : This refers to how the text is positioned in relation to the margins of the document.

- Text Highlight Color : This places a coloured rectangle over the text.
- Change Case : This allows you to change the text to uppercase, lowercase, and other styles.

Now, let's learn about some more advanced formatting commands.

Format Painter

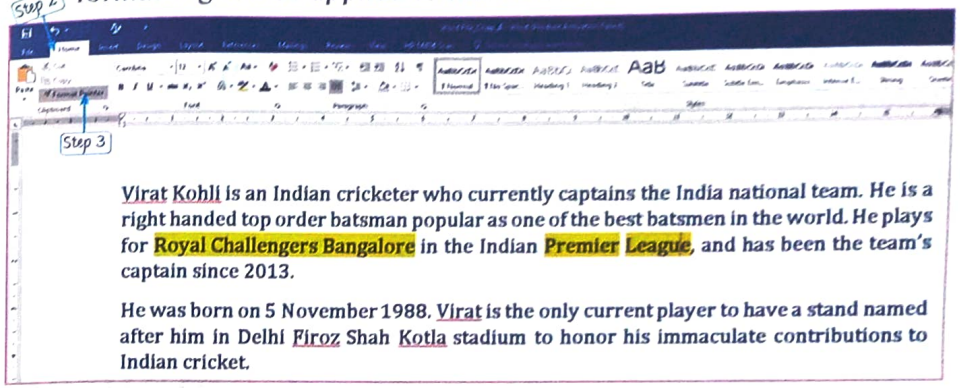
The Format Painter tool is used to copy the formatting (like font color, style, or size) from one part of your text to another. Here's how to use it :

Step 1. Select the text with the formatting you want to copy.

Step 2. Go to the Home tab on the ribbon.

Step 3. Click on the Format Painter in the Clipboard group. Your pointer will change to a brush shape (🖌).

Step 4. Click on the text where you want to apply the copied formatting. The formatting will be applied to the new text.



Superscript and Subscript

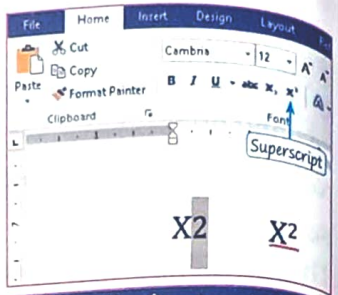
When you need to write formulas or equations in a document, you can use the Superscript and Subscript commands. Superscript places the selected text slightly above the normal text line.

To make text superscript :

Step 1. Select the text or number you want to change.

Step 2. Go to the Home tab on the ribbon.

Step 3. Click on the Superscript command in the Font group. The selected text will appear as superscript.

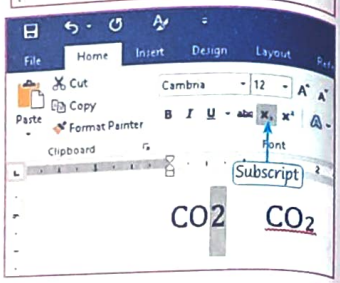


Subscript places the selected text slightly below the normal text line. To make text subscript :

Step 1. Select the text or number you want to change.

Step 2. Go to the Home tab on the ribbon.

Step 3. Click on the Subscript command in the Font group. The selected text will now appear as subscript.



Shadow Effect

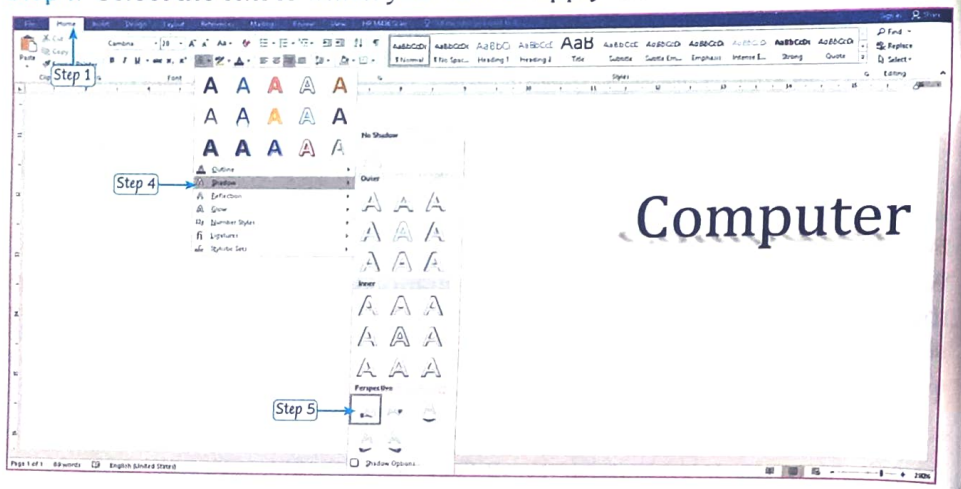
The Shadow Effect gives your text a more attractive, 3-D appearance. Here's how to add a shadow effect to your text :

SHORTCUT SECTION

To change the text into superscript, press Ctrl + Shift + =

To change the text into subscript, press Ctrl + =

Step 1. Select the text to which you want to apply the effect.



Step 2. Go to the Home tab on the ribbon.

Step 3. Click on the Text Effect command in the Font group.

Step 4. A drop-down menu will appear. Choose the Shadow option.

Step 5. Click on the Shadow Effect you want to apply to your text.

Border

You can make your document look more attractive by adding a border around a block of text or the entire page.

To add a border to selected text :

Step 1. Select the text to which you want to add a border.

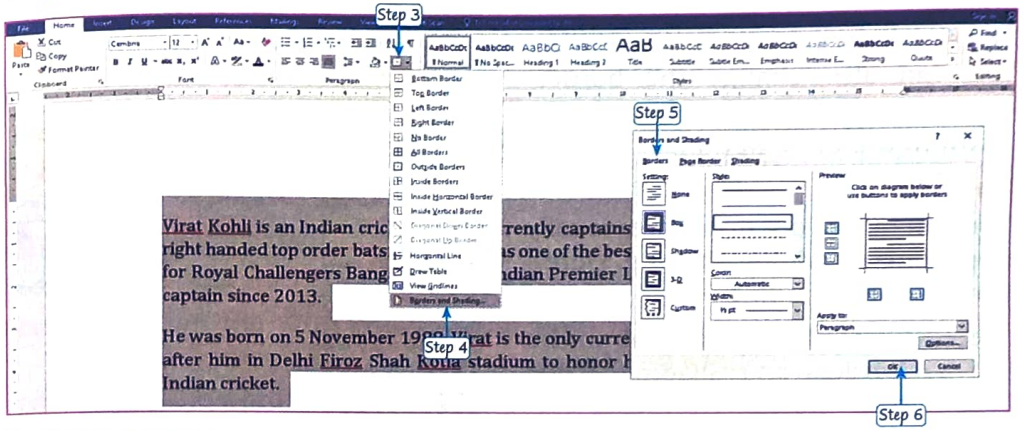
Step 2. Go to the Home tab on the ribbon.

Step 3. Click on the Borders option in the Paragraph group.

Step 4. A drop-down menu will appear. Choose Borders and Shading.

Step 5. In the Borders and Shading dialog box, go to the Borders tab. Select a border style and colour.

Step 6. Click OK button to apply the border.



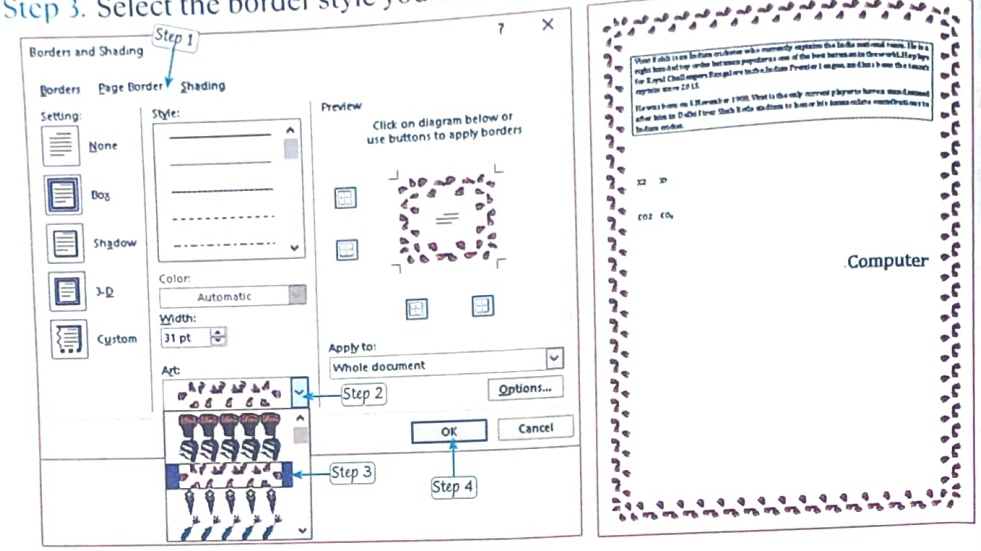
Artistic Border

To make your page more attractive, you can add an artistic border. Here's how to do it :

Step 1. In the Borders and Shading dialog box, go to the Page Border tab.

Step 2. Click the drop-down arrow next to the Art option.

Step 3. Select the border style you like.



Step 4. Click OK to apply the border.

You can also change the border colour by clicking the drop-down arrow in the Color option.

Vijay's teacher asked him to make an invitation card in Word by applying borders to it. Which group and command will he use to do so?

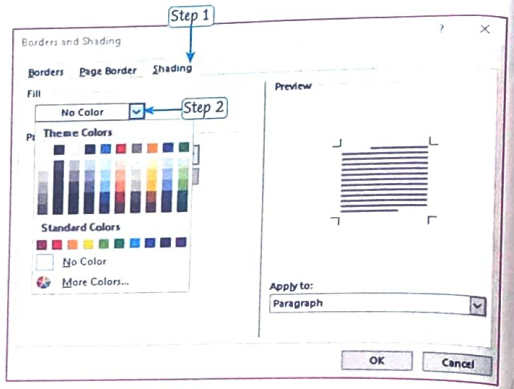
Shading

You can make your text more visually appealing by adding shading. Here's how to apply shading :

Step 1. Go to the Shading tab in the Borders and Shading dialog box.

Step 2. Click the drop-down arrow next to the Fill option.

Step 3. Choose a colour from the color palette.



Step 4. Click OK to apply the shading.

TEXT EDITING TOOLS

Editing means making changes to an existing document. You've already learned about some text editing commands :

- Selecting Text : To make changes, first select the text. You can do this by using different key combinations or by dragging the mouse over the text.
- Removing Text : Remove selected text by pressing the Backspace or Delete key.
- Cutting Text : Cutting text removes it from its original location and moves it to another location where you paste it using Ctrl + X and Ctrl + V keys.
- Copying Text : Copying text leaves the original text in place and pastes a copy elsewhere using Ctrl + C and Ctrl + V keys.
- Undo and Redo : Undo cancels the last action using Ctrl + Z, while Redo reverses the Undo action using Ctrl + Y.
- Spelling and Grammar Check : Word highlights spelling mistakes with a red wavy line and grammatical errors with a green wavy line.
- Thesaurus : The Thesaurus helps you find synonyms for a selected word.

Find and Replace

If you need to locate a specific word or phrase in a document, you don't have to read through the entire text. Word 2016 offers a Find feature to help you locate words or phrases quickly.

To find a word or phrase :

Step 1. Go to the Home tab on the ribbon.

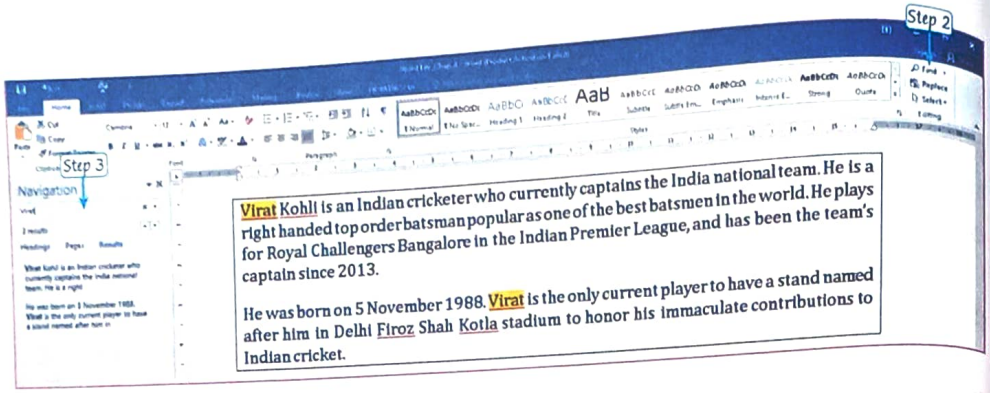
Step 2. Click on the Find command in the Editing group. This opens the Navigation panel on the left side of the Word window.

Step 3. Type the word or phrase in the text box.

Step 4. Click the Search icon or press the Enter key.

Step 5. The Navigation panel will show a list of all occurrences of the word or phrase.

Step 6. Press Enter to jump to the desired occurrence.



Replace

Word 2016 also allows you to replace a word or phrase with another throughout the document or only in selected parts.

To replace a word or phrase :

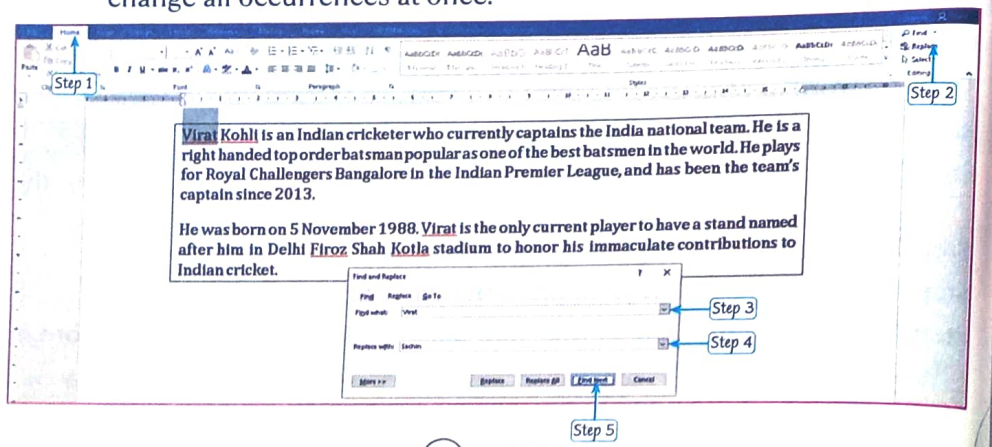
Step 1. Go to the Home tab on the ribbon.

Step 2. Click on the Replace command in the Editing group. This opens the Find and Replace dialog box.

Step 3. Type the word or phrase you want to change in the Find what text box.

Step 4. Type the new word or phrase in the Replace with text box.

Step 5. Click Find Next and Replace to selectively replace text, or Replace All to change all occurrences at once.



PAGE FORMATTING TOOLS

Changing the layout of a page is called page formatting, and it is done in the Page Setup group. Let's learn about some of the page formatting commands.

SHORTCUT SECTION
To use Find feature, press Ctrl + F
To use Replace feature, press Ctrl + H

Header and Footer

- Header is the text that appears at the top of each page.
- Footer is the text, like page numbers, that appears at the bottom of each page.

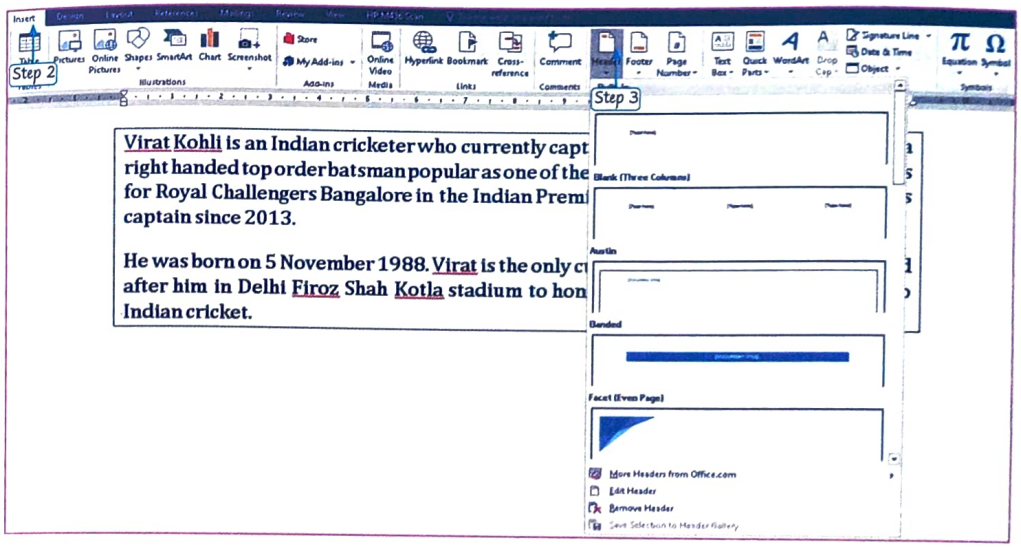
Using headers and footers saves time and effort by automatically placing the same text on each page, reducing the chances of missing important details.

Here's how to insert a header or footer in Word 2016 :

Step 1. Open your new word document in Word 2016.

Step 2. Click on the Insert tab on the ribbon.

Step 3. Find the Header and Footer options. Click on the one you need to see a list of options.



Step 4. Choose a template from the available header or footer options, or select the Blank option to type your own text. This will open the Header or Footer section on the page, where you can enter the text you want to appear on every page.

In the footer options, Word 2016 also lets you automatically add page numbers to each page.

Column and Column Break

By default, a Word document is written in a single column, but you can add more columns if needed.

To Insert Columns :

Step 1. Select the text you want to divide into columns.

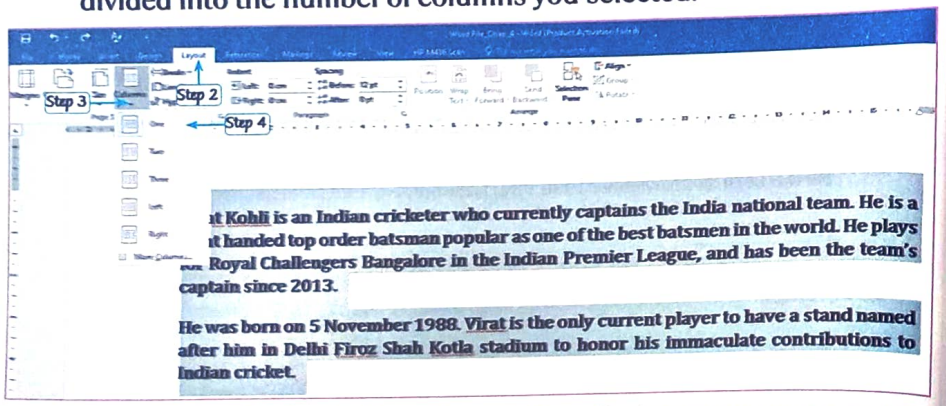
Step 2. Go to the Layout tab on the ribbon.

Step 3. Click on the Columns command in the Page Setup group.

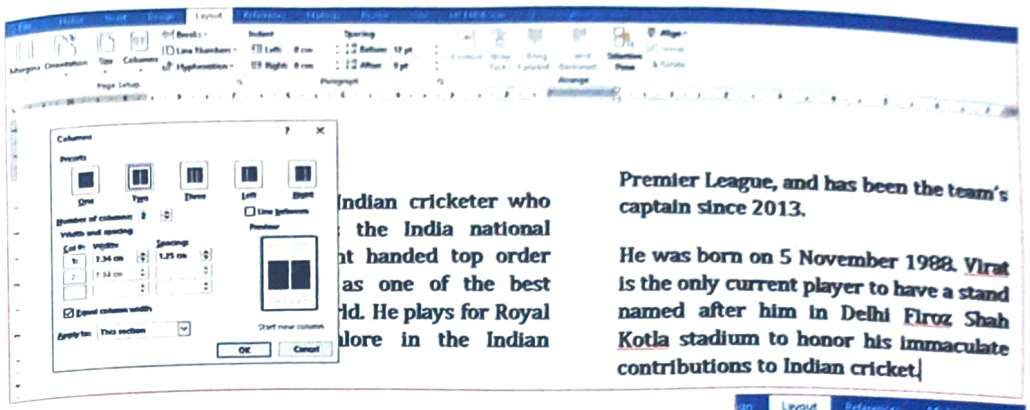
Step 4. A drop-down menu will appear. Choose the number of columns you want.

Step 5. Click on the more columns option. A dialog box will appear on the screen.

Step 6. Select the number of columns you want. The selected text will be divided into the number of columns you selected.



The selected text will be divided into the number of columns you selected. You can also add columns by clicking on the More Columns option.



Column Break

After adding columns, you may want to move text from one column to another. To do this, you can insert a column break.

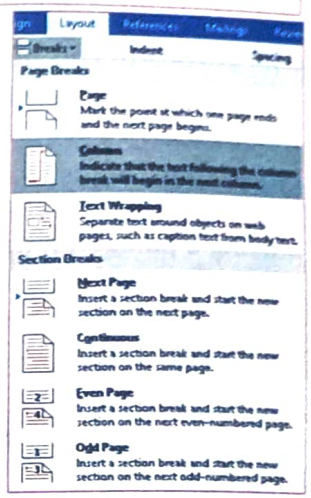
To Insert a Column Break :

Step 1. Go to the Layout tab on the ribbon.

Step 2. Place the cursor where you want to insert the column break.

Step 3. Click on the Break command in the Page Setup group.

Step 4. A drop-down menu will appear. Select the Column option.



The column break will be inserted, moving the text to the next column.

SHORTCUT To insert a column break, press Ctrl + Shift + Enter

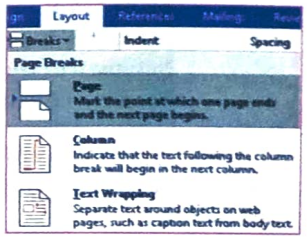
Page Break

If you want to move the text to the next page, you need to insert a page break. To Insert a Page Break :

Step 1. Go to the Layout tab on the ribbon.

Step 2. Place the cursor where you want the new page to start.

Step 3. Click on the Break command in the Page Setup group. A drop-down menu will appear.



Step 4. Select the Page option from the menu.

The page break will be inserted, moving the text to the next page.

Line Break

Pressing the Enter key starts a new paragraph, but if you want to start a new line without adding a new paragraph, use a line break. To Insert a Line Break :

Step 1. Place the cursor where you want the line break.

Step 2. Press Shift + Enter.

The text will move to a new line without starting a new paragraph.

Tabs

Tabs move the cursor half an inch by default when you press the Tab key on your keyboard. This movement is called a Tab stop. For example, pressing Tab at the start of a paragraph will indent the first line by half an inch.

Indentation

Indentation adjusts the distance of text from the left or right margin, either moving a paragraph or a single line. To Indent Text :

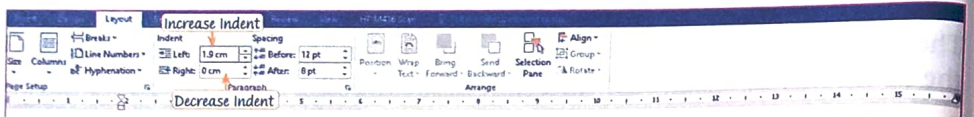
INFO Byte

If you can't see the ruler, go to the View tab and check the box next to Ruler.

Step 1. Go to the Layout tab on the ribbon.

Step 2. Select the text you want to indent.

Step 3. Click the Increase Indent button in the Paragraph group.



Virat Kohli is an Indian cricketer who currently captains the India national team. He is a right handed top order batsman popular as one of the best batsmen in the world. He plays for Royal Challengers Bangalore in the Indian Premier League, and has been the team's captain since 2013.

He was born on 5 November 1988. Virat is the only current player to have a stand named after him in Delhi Firoz Shah Kotla stadium to honor his immaculate contributions to Indian cricket.

The selected text will move to the right. To move the text back, click the Decrease Indent button.

Page Margin

Page margins are the blank spaces around the edges of the paper. These margins prevent text from being cut off when the pages are bound together in a book.

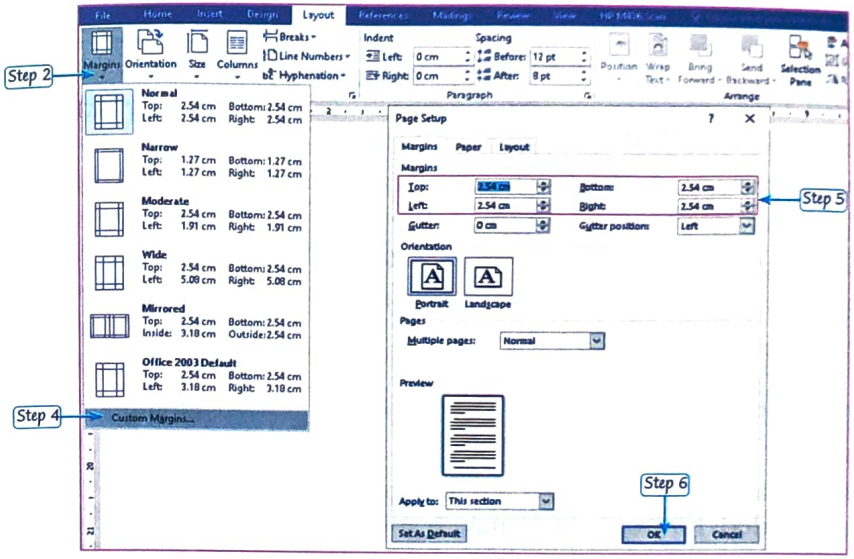
To Set Page Margins :

Step 1. Go to the Layout tab on the ribbon.

Step 2. Click on the Margins command in the Page Setup group.

Step 3. Choose a margin style from the drop-down menu.

Step 4. Select Custom Margins if you want to set your own margins. This will open the Page Setup dialog box.



Step 5. Adjust the Top, Bottom, Left, and Right margins as needed.

Step 6. Click OK to apply the changes.

Page Orientation

Page orientation determines whether the paper is oriented with the shorter edge at the top or the longer edge at the top.

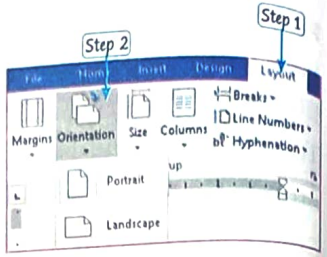
- **Portrait :** The shorter edge is at the top.
- **Landscape :** The longer edge is at the top.

To Change Page Orientation :

Step 1. Go to the Layout tab on the ribbon.

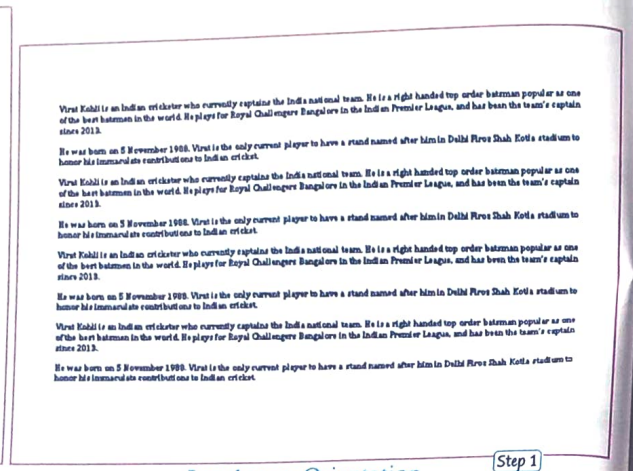
Step 2. Click on the Orientation command in the Page Setup group.

Step 3. Select Portrait or Landscape depending on your needs.



By default, Word documents are in Portrait orientation.

Portrait Orientation



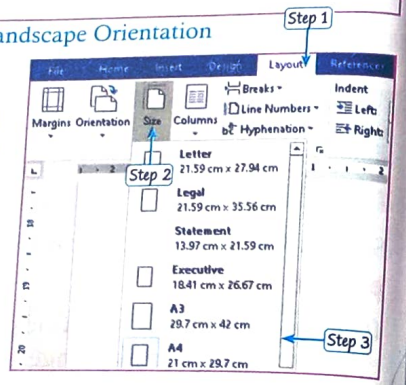
Landscape Orientation

Page Size

Word 2016 has a default page size, but you can change it if needed, for example, to match your printer's paper size.

To Change Page Size :

Step 1. Go to the Layout tab on the ribbon.



Step 2. Click on the Size command in the Page Setup group. A drop-down menu will appear showing the current page size.

Step 3. Choose the page size you want from the list.

I. Tick (✓) the correct option :

1. A header is printed in the ..... margin.
   (a) top ☐ (b) side ☐
   (c) bottom ☐ (d) middle ☐
2. You can change the space between ..... .
   (a) header ☐ (b) lines ☐
   (c) both (a) and (b) ☐ (d) None of these ☐
3. In which tab, Size command is available?
   (a) Design ☐ (b) Layout ☐
   (c) View ☐ (d) None of these ☐
4. Which of these commands places the selected text slightly above the baseline?
   (a) Subscript ☐ (b) Indent ☐
   (c) Superscript ☐ (d) None of these ☐
5. Which of these is the white space all around the printed area of the paper?
   (a) Page orientation ☐ (b) Page size ☐
   (c) Page margin ☐ (d) All of these ☐

II. Fill in the blanks :

1. By default, the orientation of a Word document is ..... .
2. Page break option is present under the ..... tab.
3. The ..... option places the selected text slightly below the baseline.
4. The ..... is places at the end of a document.

III. Write 'T' for true and 'F' for false :

1. Paragraph spacing cannot be changed in a document.
2. Header is placed at the bottom of a page.
3. To make changes in the page layout is called page formatting.
4. Page break is used to align the text.
5. Shadow effect makes the text more attractive and looks like 3-D text.

IV. Answer the following questions :

1. Define footer.
2. Name the types of page orientation in Word.
3. How do you use the Find and Replace feature of Word?
4. What are the Tabs? Write the default settings of the tab.

Smart Thinking

❖ **Complete the given steps to apply shadow to the text.**

1. Select the ..... .
2. Click on ..... .
3. Click on ..... command in the ..... group.
4. Select option ..... from the drop-down menu.
5. Click on any ..... .

❖ **Find 6 term related to word processor.**

T	G	H	K	S	D	I	P
I	S	N	E	A	K	N	C
F	O	O	T	E	R	D	O
D	T	A	B	S	J	E	L
G	V	X	B	R	T	N	U
N	L	I	N	E	N	T	M
T	V	O	N	E	A	E	N
H	E	A	D	E	R	T	E

Lab Activity

AIM : Open Word in your computer and do these things :

- Create a document in Word and type two of the poems of your textbook.
- Write each poem in different style, bold the name of the poem.
- Type the following by using subscript/superscript command :
  (a) $x^4 - x^3 + 3x^2 + 2x + 2$
  (b) $12x^3 - 2x^2 + x + 1$
  (c) SO_2
  (d) CO_2
- Write the following text in the landscape orientation. Use different format commands like Bold, Underline and Italic.

(a) **Software**

Software is a set of instruction that makes the computer perform tasks. It controls the computer and makes it do useful work.

Types of software :

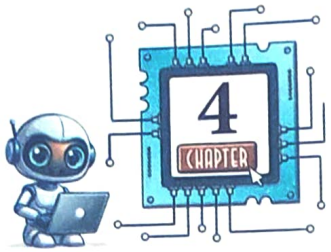
- System Software
- Application Software

(b) To make changes in existing documents is called editing. The tools used to edit documents are called editing tools. **MS Word** offers various editing tools : Grammar checker, copy, cut and paste, redo and undo, Select and delete, etc.

Superscript command places the selected text slightly above the base line. **Subscript** command places the text slightly below the base time.

FOR THE TEACHER

Provide the students with unformatted text and instruct them to create a chapter by adding heading, headers, footers and other features of a word processor that were discussed in the chapter.



PowerPoint 2016

Let's Recall

Which of the following icons is not an application of Microsoft?

In this Chapter

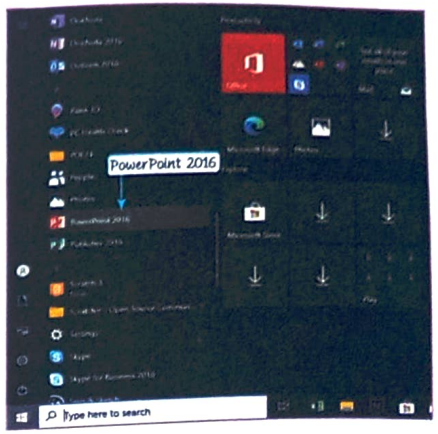
- Starting PowerPoint 2016
- Creating a new Presentation
- Opening a Saved Presentation
- Exiting PowerPoint
- Components of the PowerPoint Window
- Saving a Presentation
- Closing a Presentation

PowerPoint is a software program that helps you create and display slides for presentations. You can add text, images, and multimedia to make your presentations look professional. With PowerPoint, you can:

- Organise and structure your presentation
- Create a consistent and polished look
- Add visual interest to your presentation with animations

It's a popular tool among teachers, students, and professionals for making appealing presentations. You can also use PowerPoint to create handouts for your audience, outlines, and speaker notes.

STARTING POWERPOINT 2016

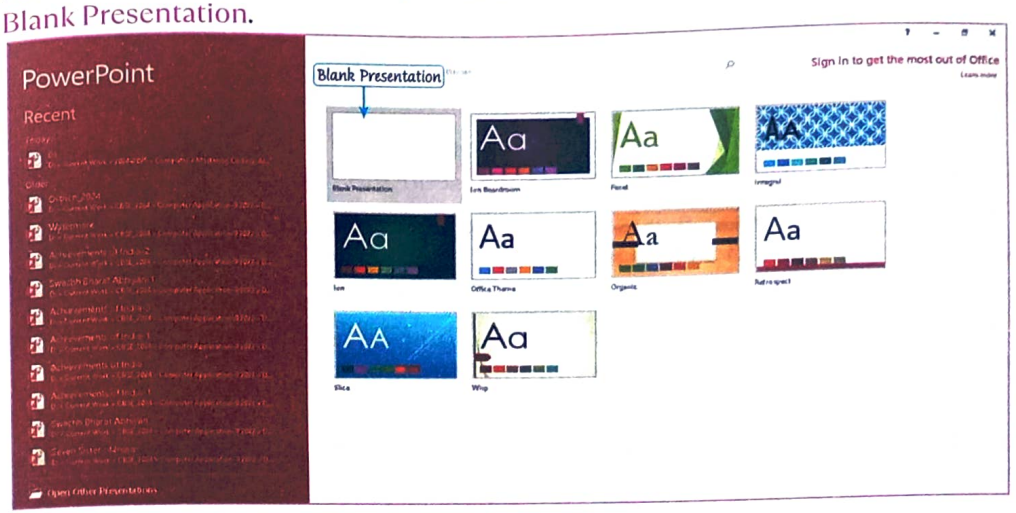


To open PowerPoint, follow these steps :

Step 1. Click the Start button.

Step 2. Scroll down and click on PowerPoint or PowerPoint 2016.

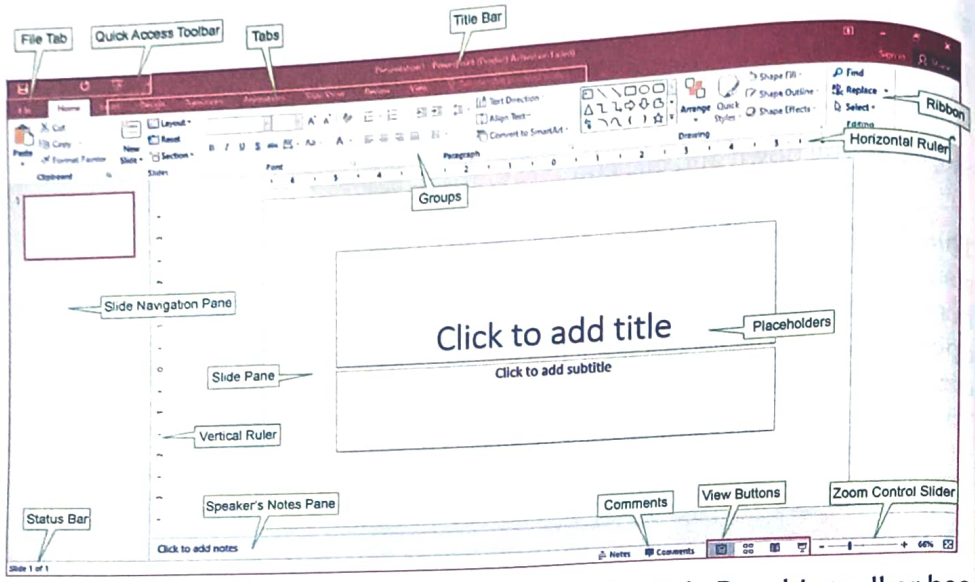
This will open the PowerPoint Start screen, also known as the Backstage view. From here, you can create new presentations, open existing ones, save your work, print, create PDFs, and share your presentations. To start a new presentation, just click on Blank Presentation.



COMPONENTS OF THE POWERPOINT WINDOW

Here's a simple overview of the main parts of the PowerPoint 2016 window :

Title Bar : Located at the top of the window, the title bar shows the name of your current presentation or the name you saved it under. It also displays the program name. On the right side, there are three buttons : Minimize, Maximize/ Restore Down and Close.



Quick Access Toolbar : Found on the left side of the Title Bar, this toolbar has commonly used buttons like Save, Undo, Redo and Slideshow.

Ribbon : The Ribbon consists of Tabs and Groups. Each tab contains a set of commands, displayed as tool buttons.

Slides/Outline Pane : On the left side of the window, this pane shows thumbnails of each slide in your presentation (Slides Pane) or a text outline of all the information on the slides (Outline Pane).

Speaker's Notes Pane : This area allows the speaker to write notes for themselves. These notes are only visible to the speaker and don't appear during the presentation.

Placeholder : A placeholder is a dotted box on a slide where you can add text, pictures, videos, etc.

Status Bar : Located at the bottom of the window, the status bar shows details like the current slide number, total number of slides, and the design theme used.

CREATING A NEW PRESENTATION

A presentation is made up of multiple slides, where each slide is like a single page. A slide can contain text, animations, videos, sound effects, and voiceovers. When these slides are shown one after another, it's called a slideshow. When you open PowerPoint, it automatically starts a new presentation called Presentation 1.

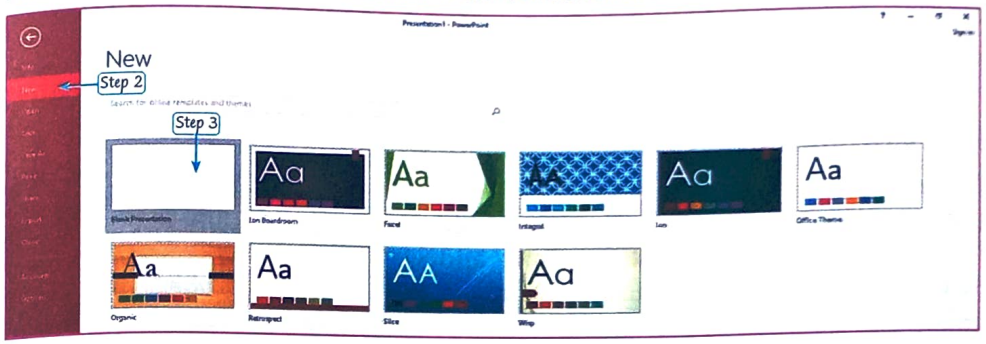
To create a new presentation, follow these steps :

Step 1. Click the File tab.

Step 2. Click on New option.

Step 3. In the Backstage View, click on Blank Presentation.

This will open a new presentation with a blank slide.



Creating the Title Slide

When you start your presentation, the first slide that appears is the Title slide by default. This slide has a text placeholder where you can type your title.

To enter text in the Title Text Placeholder :

Step 1. Click on the title text placeholder box.

Step 2. Type your text in the box. For example, in the image, the text entered is I LOVE MY COUNTRY.

The title you type will appear in the placeholder box and also in the Slides tab.

Entering Text in the Subtitle Text Placeholder :

After adding your title, the next step is to enter text in the subtitle text placeholder. To do this :

Step 1. Click on the subtitle text placeholder box.

Step 2. Type your text in the box.

The text you enter will appear in the subtitle placeholder and in the Slides tab.

Inserting a New Slide

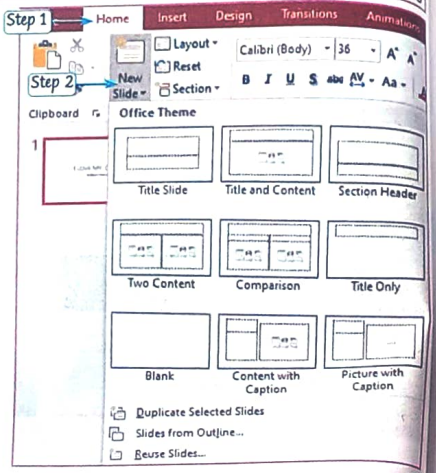
To add a new slide, follow these steps :

Step 1. Click on the Home tab.

Step 2. Click on the New Slide button in the Slides group.

Step 3. Choose the type of slide you want.

A new slide will be added to your presentation. You can also add a new slide by right-clicking in the Slides Pane and selecting New Slide from the pop-up menu.



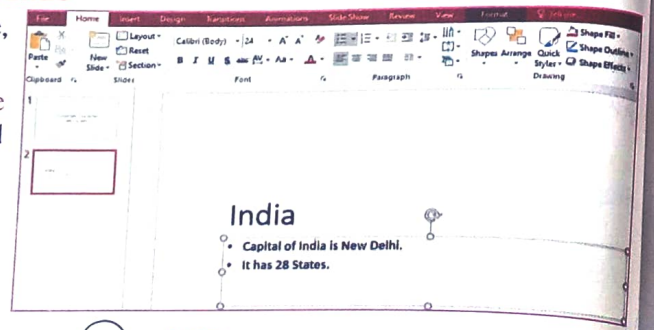
SHORTCUT SECTION To add new slide, press Ctrl + M

Entering Text in the New Slide

To add text to the new slide, follow these steps :

Step 1. Click on the title text placeholder and type your text.

Step 2. Click on the subtitle text placeholder and type your text. Press Enter to start a new bullet point on the next line and continue typing your text.



Changing the Font and Font Size of the Text

You can change the font and size of your slide text by following these steps :

Step 1. Select the text you want to change.

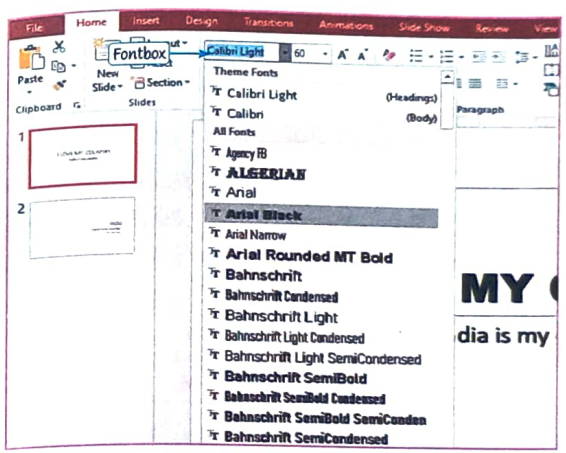
Step 2. Click on the Home tab.

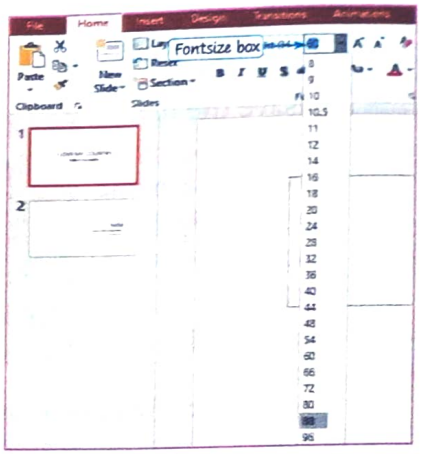
Step 3. In the Font group, click the drop-down arrow in the Font box to see the available fonts, or click the down arrow in the Font Size box to see the available sizes.

Step 4. Choose the font type and size you want for your text.

Let's Get Better

Rohit's teacher asked him to create a PowerPoint presentation of 10 slides. However, he forgot how to add slides to his presentation. Suggest him a method to do so.





Activity Corner

Rearrange the letters to get correct word :

1. OWPEROPINT
2. BCAKSATEG
3. RNESEPNOITAT

4. LIDES

5. PCELOLDHERA

SAVING A PRESENTATION

You can save your presentation to share with others or use later by following these steps :

Step 1. Click on the File tab.

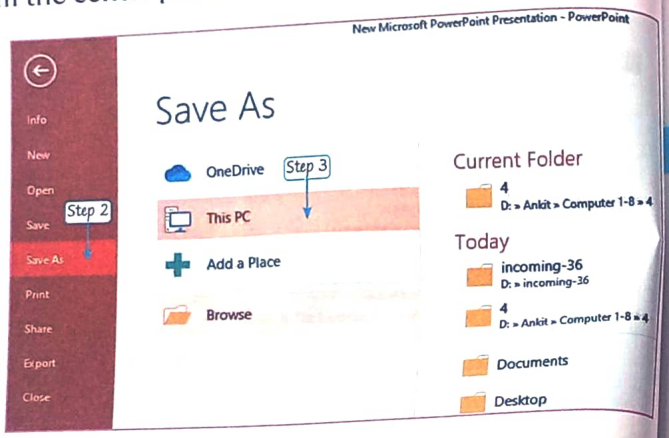
Step 2. Choose Save or Save As from the left pane. You can also click the Save button on the Quick Access Toolbar.

Step 3. Select This PC from the center pane, then click on Browse.

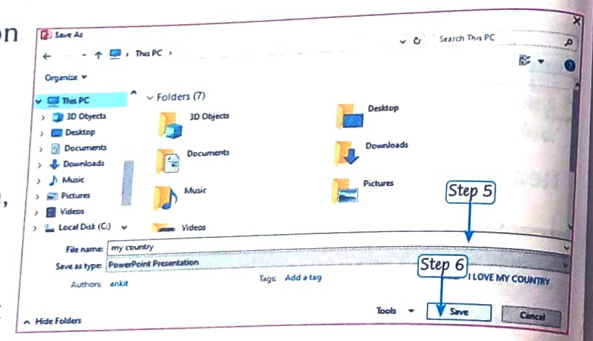
Step 4. The Save As dialog box will open. Choose where you want to save the file.

Step 5. In the File name box, type a name for your file.

Step 6. Click the Save button.



PowerPoint will save your presentation, and the new file name will appear on the title bar.

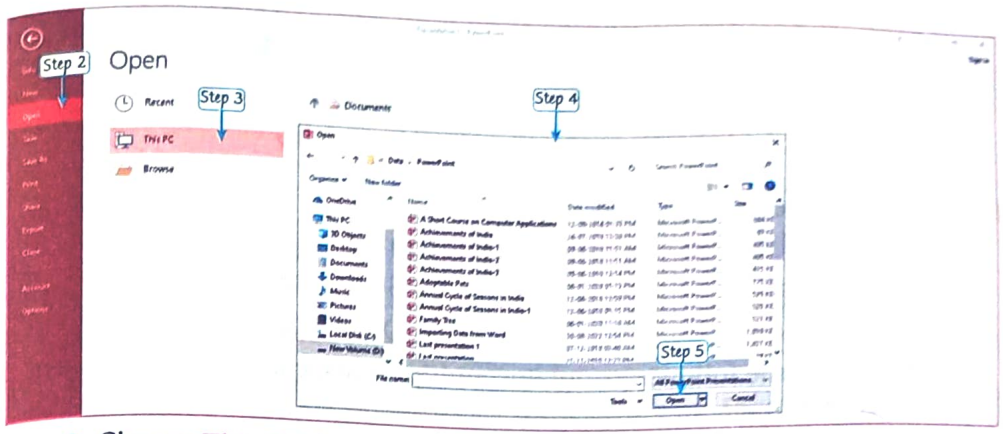


OPENING A SAVED PRESENTATION

To open a saved presentation, follow these steps :

Step 1. Click on the File tab.

Step 2. Select Open from the left pane.



Step 3. Choose This PC from the centre pane, then click Browse.

Step 4. The Open dialog box will appear. Find the location of your file.

Step 5. Click the Open button.

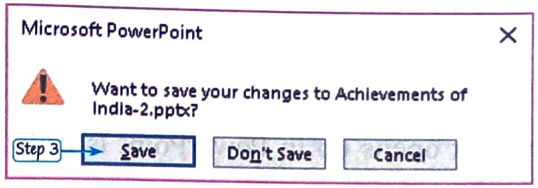
CLOSING A PRESENTATION

To close the presentation, follow these steps :

Step 1. Click on the File tab.

Step 2. Select Close from the left pane. If you haven't saved your work, PowerPoint will ask if you want to save your changes.

Step 3. Click Save if you want to save the presentation, or click Don't Save if you don't want to save it.



PowerPoint will then close the presentation.

EXITING POWERPOINT

To exit PowerPoint, click the Close button × in the Title Bar.

SHORTCUT SECTION To quit PowerPoint, press Ctrl + Q

I. Tick (✓) the correct option :

1. Name of the presentation is displayed on the ..... .
   (a) Title bar ☐ (b) Status bar ☐
   (c) Taskbar ☐ (d) Quick Access Toolbar ☐
2. Which of the following contains multiple tabs, each with several group of commands?
   (a) Status bar ☐ (b) Slide pane ☐
   (c) Ribbon ☐ (d) Placeholder ☐
3. Which of the following displays the slides as thumbnails?
   (a) Outline pane ☐ (b) Slides pane ☐
   (c) Work area ☐ (d) Ribbon ☐
4. In which tab, Save As option is present?
   (a) Insert ☐ (b) File ☐
   (c) Format ☐ (d) View ☐
5. Which of the following is the shortcut keys to insert a new slide?
   (a) Crtl + M ☐ (b) Alt + M ☐
   (c) Shift + M ☐ (d) Shift + N ☐

II. Fill in the blanks :

1. The first window that opens up in PowerPoint is called ..... view.
2. ..... is located at the top of the PowerPoint window.
3. A single page of a presentation is called a ..... .
4. ..... button is used to exit the PowerPoint.

III. Write 'T' for true and 'F' for false :

1. To close PowerPoint, you can click on Minimize button. ☐
2. Quick Access Toolbar is located at the top right corner of the taskbar. ☐
3. A presentation is made up of a series of slides. ☐
4. You cannot change the font type of the text in slides. ☐
5. The content of the presentation is inserted in text placeholder. ☐

IV. Answer the following questions.

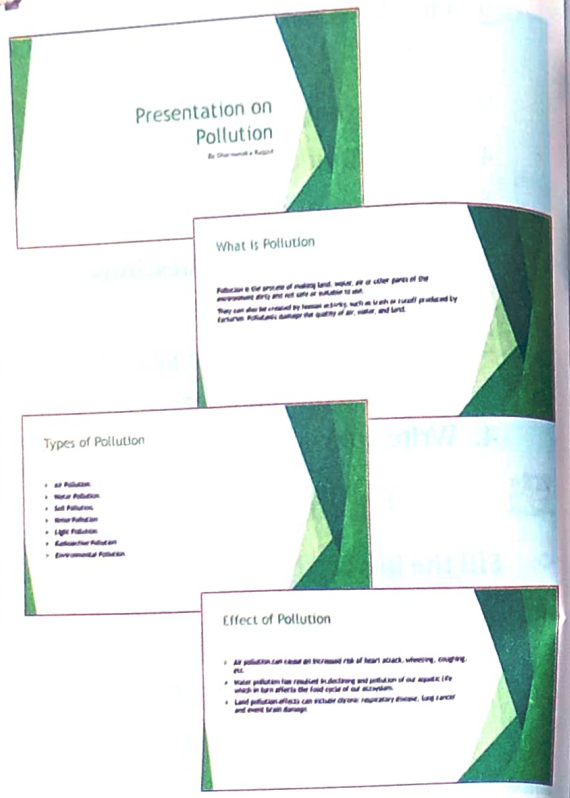
1. What is PowerPoint?
2. What is a slide? Write steps to create Title slide.
3. What is a placeholder?
4. Write steps to save presentation.

❖ **Fill the boxes given below as instructed :**

Number	10 More	10 Less	100 More	100 Less
133				
467				
733				
947				
569				
799				
667				
113				

Lab Activity

1. Open PowerPoint to create a presentation on 'Pollution'. Give the information about the types and causes of pollution, the importance and need of plants, effects of cutting plants, etc. in atleast 3 slides.
2. Create some slides to show ways to prevent pollution. Format the text using attractive fonts, size and colour.
3. Save your presentation with the name 'Pollution' and close PowerPoint.



FOR THE TEACHER

Demonstrate a presentation with special effects to give the students an idea of how effective a presentation can be.

5 CHAPTER

More on PowerPoint

Let's Recall

Which of the following is the icon of Microsoft PowerPoint?

In this Chapter

- Slide Layout
- Viewing a Presentation
- Inserting WordArt
- Deleting a Slide
- Inserting a Picture from a File

In the previous chapter, we learned about Microsoft PowerPoint. You already know how to create a slide and add text to it. In this chapter, you'll explore some advanced features of PowerPoint.

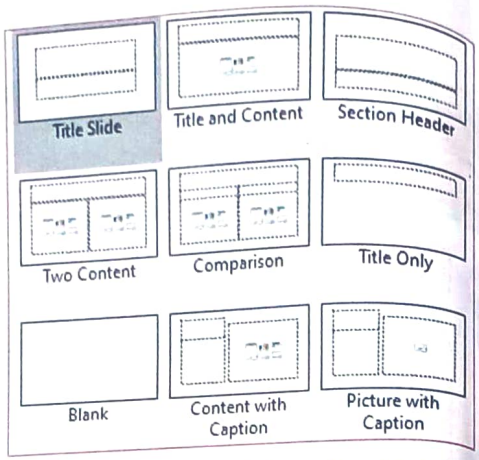
SLIDE LAYOUT

A slide layout refers to how the content boxes, called placeholders, are arranged on a slide. These placeholders are the dotted-line boxes that hold various types of content, such as titles, body text, tables, SmartArt, charts, pictures, ClipArt, videos and sounds.

Slide layouts also include the theme of the slide, which involves the colors, fonts, effects, and background. Now, let's learn more about slide layouts and how you can change them on a slide.

Types of Built-in Slide Layouts

There are nine built-in slide layouts in PowerPoint. Let's look at these layouts :



Title Slide : This layout is used at the start of your presentation to introduce the topic.

Title and Content : This is the default layout and is commonly used. It includes space for a title and content.

Section Header : This layout is used to separate different sections within the same presentation.

Two Content : This layout has two columns for text and graphics, allowing you to display two sets of content side by side.

Comparison : Similar to the Two Content layout, but with a heading text box above each column. This layout is great for comparing two types of content, like different charts.

Title Only : This layout only includes a title, leaving the rest of the slide blank for you to add other content like images or charts.

Blank : A blank layout is useful when you need a slide with just a picture or graphic, without additional text.

Content with Caption : This layout has two columns. The left side is for text, and the right side is for images or illustrations.

Picture with Caption : Similar to the Content with Caption layout, but the right side is specifically for a picture stored on your computer, with text on the left side.

You can also create custom layouts to fit your specific needs and share them with others who create presentations in PowerPoint.

Changing the Slide Layout

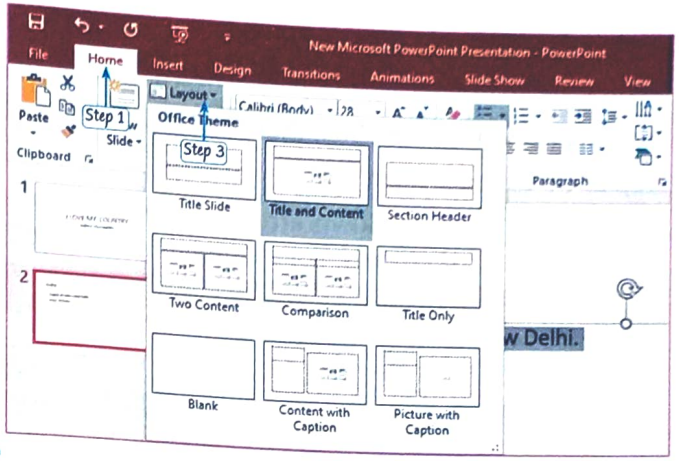
You can change the slide layout to suit your needs. To do this, follow these steps :

Step 1. Click on the Home tab.

Step 2. Select the slide you want to change.

Step 3. Click the Layout button in the Slides group.

Step 4. Choose the layout you want to use.



INSERTING WORDART

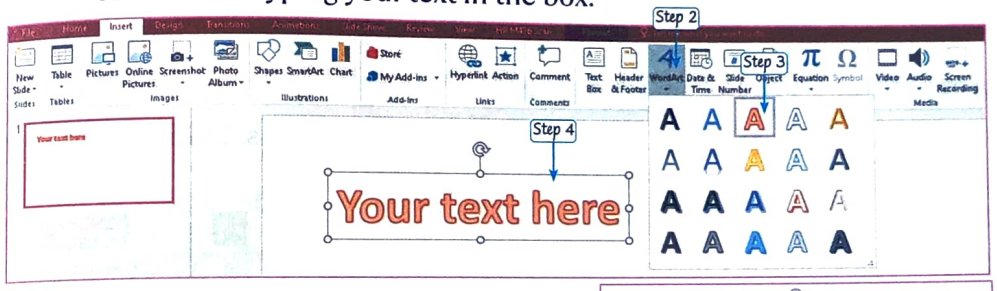
WordArt allows you create special text effects that aren't possible with regular font formatting. You can use WordArt to make your slides more visually appealing. Here's how to insert WordArt :

Step 1. Click on the Insert tab.

Step 2. In the Text group, click on the WordArt command. A drop-down menu with WordArt styles will appear.

Step 3. Choose the WordArt style you like.

Step 4. A text box will appear on the slide with the placeholder text "Your Text Here." Start typing your text in the box.



Step 5. Once you're done typing, click anywhere outside the text box or press the Escape key twice.

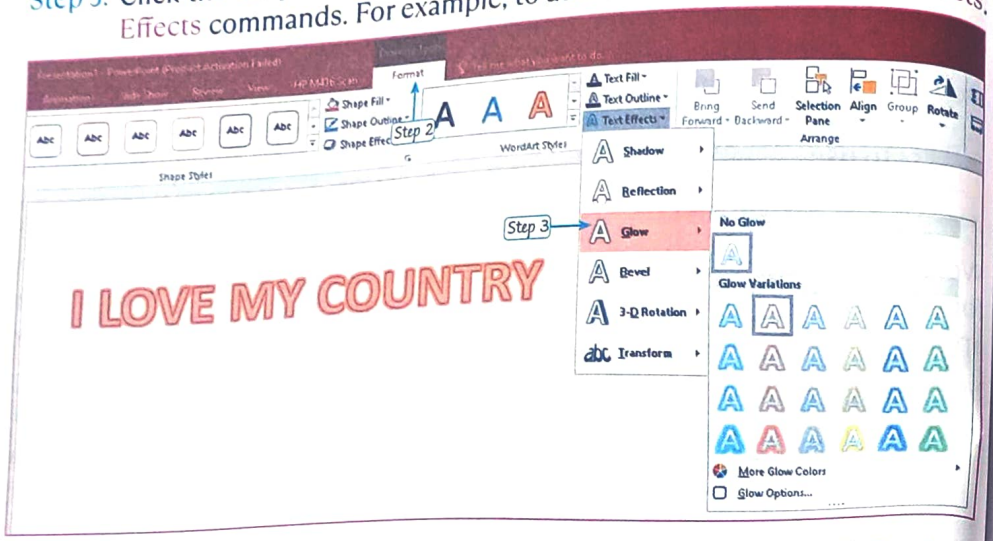
Editing the WordArt

If you want to change or edit your WordArt, follow these steps:

Step 1. Click on the text box, then triple-click the text to select it.

Step 2. Go to the Format tab.

Step 3. Click the drop-down arrow next to the Text Fill, Text Outline or Text Effects commands. For example, to add an effect, click on Text Effects.



Step 4. A menu will appear with different options like Shadow, Reflection, Glow, Bevel, 3-D Rotation, and Transform. Choose an option.

Step 5. A submenu will open. Click on the sub-option you want to apply.

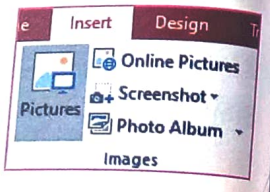
INSERTING A PICTURE FROM A FILE

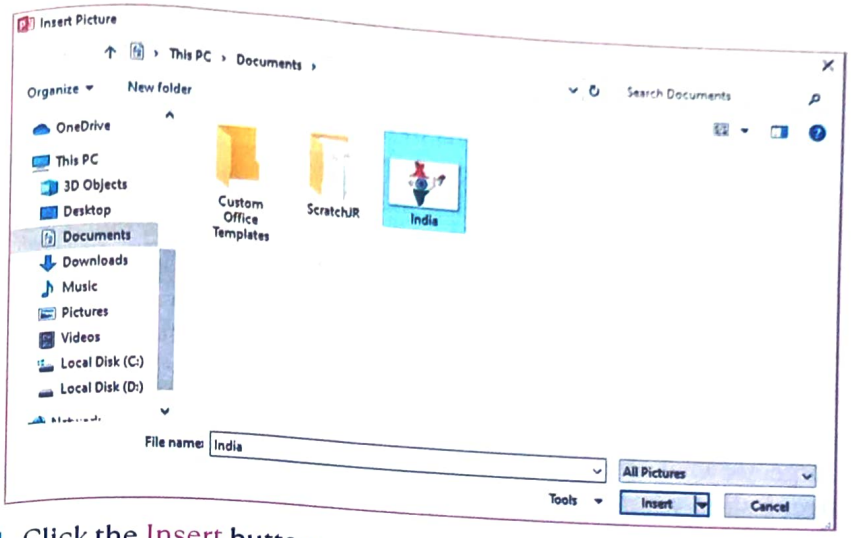
PowerPoint allows you to add images to your presentations. To insert a picture from a file, follow these steps :

Step 1. Click on the Insert tab.

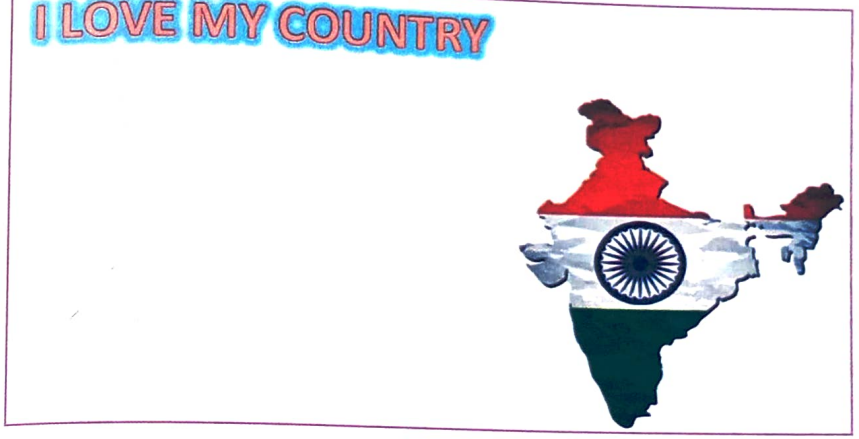
Step 2. Click on the Picture button in the Images group. The Insert Picture dialog box will appear.

Step 3. Browse to the location where your picture is saved and select the file.





Step 4. Click the Insert button.



The selected picture will be added to the slide.

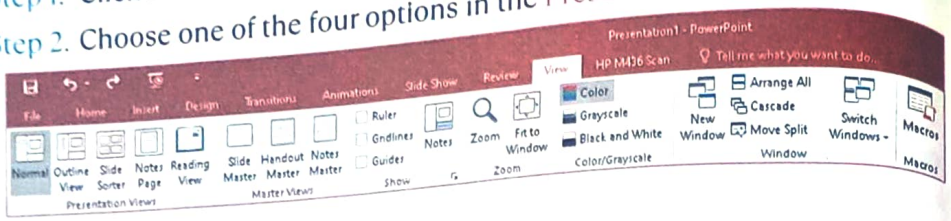
VIEWING A PRESENTATION

PowerPoint offers different view options to change how your presentation appears on the screen, helping you focus on various elements you want to edit. By default, PowerPoint opens in Normal View.

To change the view of your presentation, follow these steps :

Step 1. Click on the View tab.

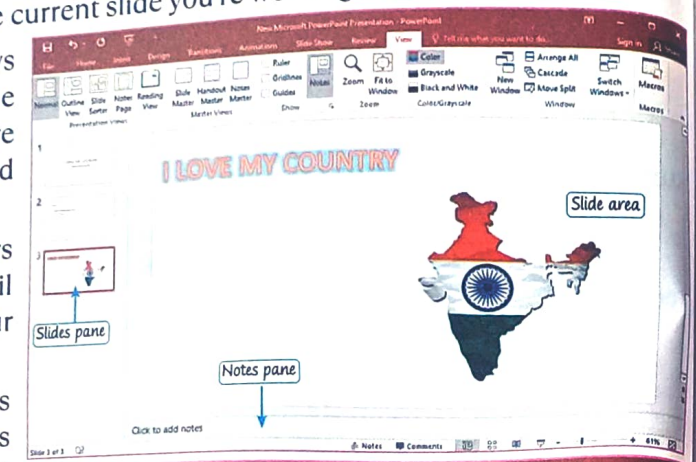
Step 2. Choose one of the four options in the Presentation Views group.



Normal View

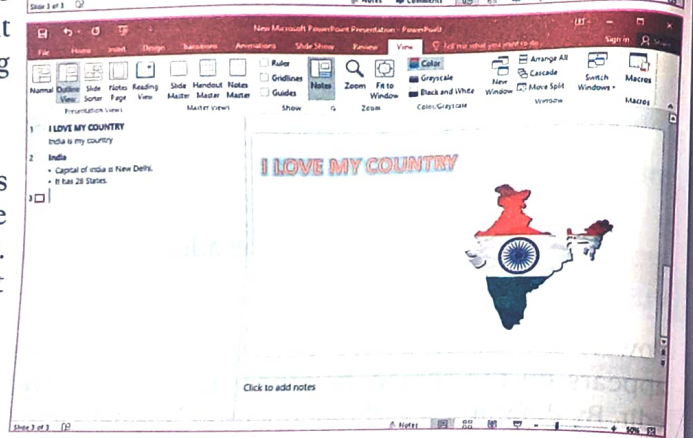
Normal View shows the current slide you're working on.

- Slide Area : Displays a large view of the current slide where you can create and edit slides.
- Slides Pane : Shows small thumbnail versions of all your slides.
- Notes Pane : Allows you to write notes for yourself that won't appear during the presentation.



Outline View

Outline View displays the text from each slide in a list on the left side. You can edit the text directly from this view.



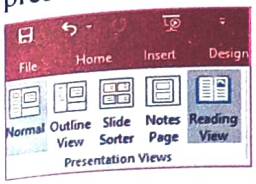
Slide Sorter View

Slide Sorter View displays miniature versions of all your slides, making it easy to rearrange their order. This view is helpful because you can see many slides at once, and you can move, delete, or edit them with ease. To switch to this view, click on the Slide Sorter View command in the Presentation Views group under the View tab.

SHORTCUT SECTION To switch between slides, press Ctrl + Shift + Tab

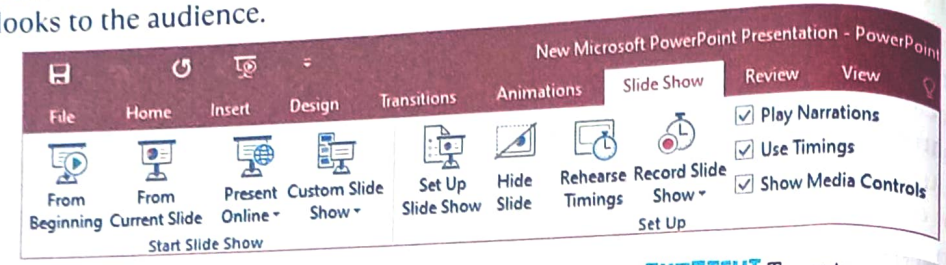
Reading View

Reading View lets you view your presentation in a window rather than full screen, with simple controls for easy review. The ribbon and slides/outline pane are hidden. To switch to this view, click on the Reading View command in the Presentation Views group under the View tab. PowerPoint shows the presentation in a window with the Title Bar, Status Bar and simple controls.



Slide Show View

Slide Show View is used when you want to present your slides to an audience. This view runs your presentation as a slide show, allowing you to see how it looks to the audience.



Changing the Slide Order

You can easily move a slide by selecting it and dragging it to a new position. If you want to rearrange the slides, you can do this in either Normal View or Slide Sorter View.

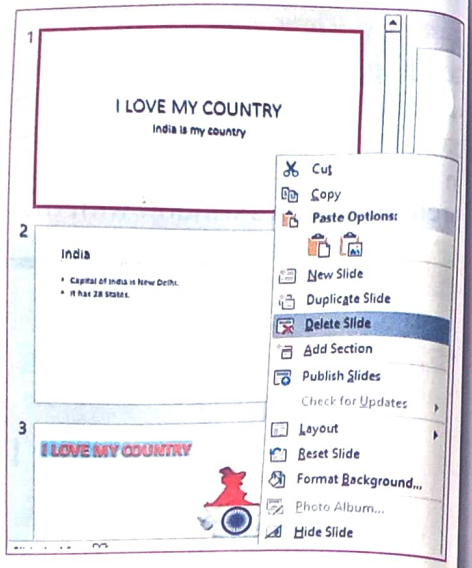
SHORTCUT SECTION To return to normal view, press Esc

Deleting a Slide

You can delete slides in either Normal View or Slide Sorter View. Here's how :

Step 1. Right-click on the slide in the Outline Pane.

Step 2. Choose Delete Slide from the menu that appears. The slide will be removed.



Let's Get Better

Arun prepared a slide on World Environment Day which consists of around 10 pages. He used Picture with caption layout. Now, he wants to display all the slides altogether. Which view he will use to do so?

SHORTCUT SECTION To undo an action, press Ctrl + Z

Activity Corner

Create a slideshow titled as 'Cosmos'.

- Insert images related to the topic.
- Use different font colour, size and style for each slide.
- Type facts you know about universe.

Exercise

I. Tick (✓) the correct option :

1. Which of these is dotted-line containers on slide layout?
   (a) Placeholders ☐ (b) Animation ☐
   (c) Clipart ☐ (d) None of these ☐
2. Which of the following allows you to create text effects that is not possible through formating?
   (a) WordArt ☐ (b) Picture ☐
   (c) Animation ☐ (d) Transitions ☐
3. Which of the following is the arrangement of text, graphics, images, etc?
   (a) Animation ☐ (b) Slide Layout ☐
   (c) WordArt ☐ (d) Slide Pane ☐
4. Which of the following shows the miniature version of the slides?
   (a) Slide View ☐ (b) Slide Sorter View ☐
   (c) Outline View ☐ (d) Reading View ☐

II. Fill in the blanks :

1. ..... view is used to deliver your presentation to the audience.
2. PowerPoint 2016 includes ..... in-built style layouts.
3. ..... displays a large view of the current slide.
4. ..... view displays full screen images of your slides.

III. Match the following :

1.
2.
3.
4.
5.

(a) Reading view
(b) Slide Show
(c) Normal view
(d) Slide Sorter view
(e) Outline view button

IV. Write 'T' for true and 'F' for false :

1. There are six different ways to view your presentation.
2. Slide show view displays miniatures of your slides.
3. You cannot change the font of the WordArt.
4. WordArt command is present under the Insert tab.

V. Answer the following questions :

1. What is slide layout? How many slide layouts are available in PowerPoint?
2. Write the difference between slide sorter view and reading view.
3. Write steps to insert WordArt in the slide.
4. Write steps to insert a picture from a file in the slide.
5. How can you delete a slide from your presentation?

Smart Thinking

❖ **Which screen view will you use in the following?**

To see the slide on the entire screen .....

To write something on the slide .....

To write a note for the speaker .....

To see the pictures of all the slides .....

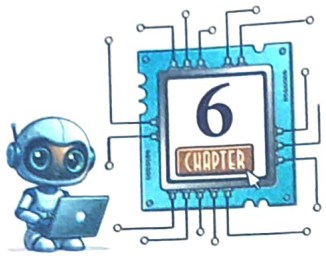
Lab Activity

Create a presentation on Air pollution or Water shortage.

- Write speaker notes.
- View your presentation in Slide Show view.
- Save your presentation with name 'Water shortage'.

FOR THE TEACHER

- Explain the benefits of using Powerpoint as a presentation tool.
- Demonstrate each element of slide layout and graphics covered in this chapter, including the use of images or WordArt in presentations.
- Show the various types of views available of Powerpoint.



Internet

Let's Recall

Which of the following is an output device?

In this Chapter

- What is the internet?
- Opening a Website
- Uses of the Internet
- Good Practices
- Basic Terms
- How to be Safe Online?

Many of you might have heard of the term, the Internet. The Internet is a big network that connects computers all over the world, allowing us to share information, talk to people, and find lots of interesting things online. In this chapter, you will learn what the Internet is, how we use it every day, and why it's important to stay safe while exploring it. Let's discover how the Internet makes our lives easier and more fun!

WHAT IS THE INTERNET?

The Internet is a big network that connects computers all around the world. These computers are linked together by telephone wires and other ways, so they can share information with each other. The Internet is also commonly known as the Net.

USES OF THE INTERNET

- The Internet is a lot of fun! We can check the latest cricket scores, look up the weather for a place we want to visit, and even read the whole newspaper online.
- We can use the Internet to book train and flight tickets, and we can also shop for things like clothes, books, and shoes.
- We can send messages, called emails, to any part of the world in just a few seconds. We can also chat with friends, family, and relatives online.

BASIC TERMS

To understand the internet, it's essential to know key terms used while browsing the web!

World Wide Web (WWW)

The World Wide Web, or the Web, is a collection of special documents called web pages that are connected to each other. You can access the Web from anywhere in the world using the Internet. To view the World Wide Web, you need something called a browser.

INFO Byte

The World Wide Web was developed in 1989 by Sir Tim Berners-Lee.

Browser

A browser is a program that helps you find and view web pages. It also lets you interact with text, pictures, videos, and music on those web pages. Some popular browsers are : Google Chrome, Opera, Safari etc.

Website

A website is a spot on the Internet where someone shares information. It contains a group of related web pages that are linked together with hyperlinks. A website is like a magazine–you can move from one page to another using these links, just like flipping through the pages of a magazine.

Going from one website to another is called net surfing or just surfing.

INFO Byte

The World Wide Web is believed to contain more than 20 billion web pages. That is 20 followed by nine zeros.

Web Page

A web page is a single page on the Internet that displays information like text, pictures, videos, and more. It's similar to a page in a book but appears on your computer or phone screen. Each web page is part of a larger collection called a website, and you can move between different web pages using links.

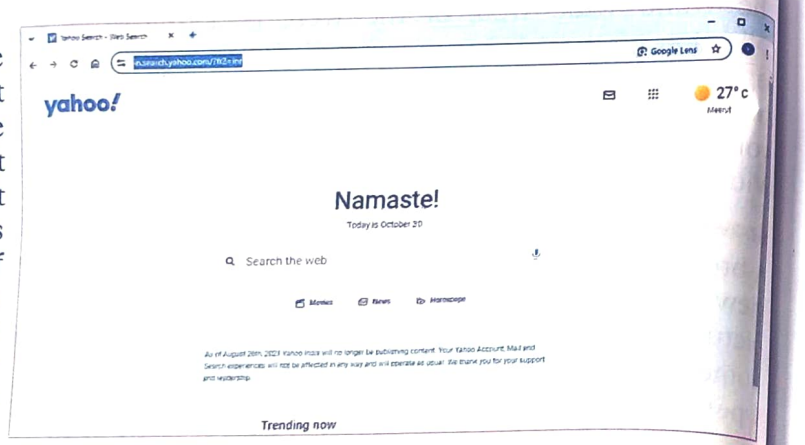
This page may have text, often underlined and coloured, or small pictures called icons. When you click on this text or these images, it opens another web page.

The link that connects one page to another is called a hyperlink. When you move your mouse pointer over a hyperlink, it changes to a hand symbol.

Hyperlinks let you jump to a different web page or move to another part of the same page.

Home Page

The home page is the first page you see when you visit a website. It usually gives you a brief introduction to the website and has links to more details about the services or information the website offers.

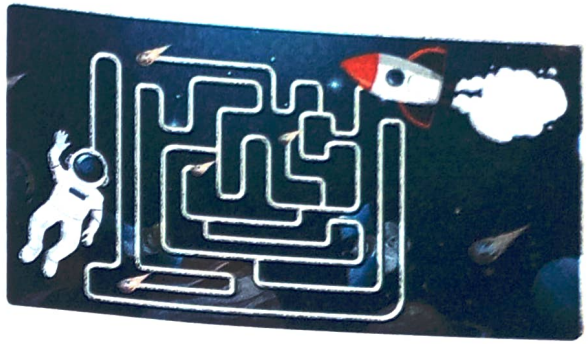


Let's Get Better

While moving through a website, Rohan noticed that the mouse pointer changed to a hand symbol when the cursor was brought near some text. What is such text called?

Activity Corner

Let's help the Astronaut get back to the Spaceship

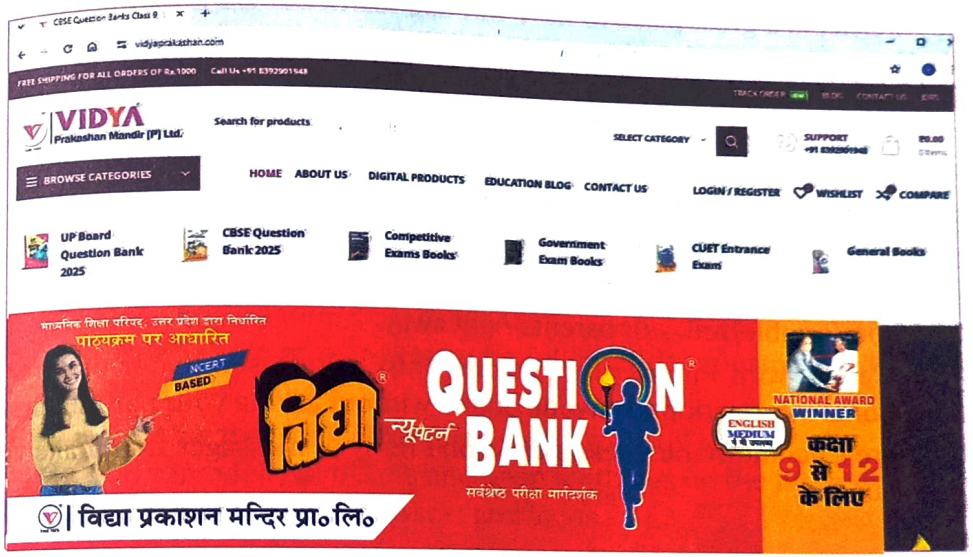


OPENING A WEBSITE USING A BROWSER

Here are the simple steps to open a website using a browser :

Step 1. Click on the browser icon on your computer or phone (like Google Chrome, Microsoft Edge, or Safari).

Step 2. In the address bar at the top of the browser, type the website's address, like 'www.vidyaprakashan.com'.



Step 3. After typing the address, press the Enter key on your keyboard.

Step 4. The browser will connect to the website, and you'll see the web page appear on your screen.

Step 5. You can now click on links and buttons to navigate around the website and find the information you're looking for.

GOOD PRACTICES

Just like in the real world, there are certain rules to follow on the Internet. These rules are called Netiquette. Here are some good practices to keep in mind :

- Don't Use All Caps : Writing in all capital letters in emails or messages can seem like shouting.
- Check Your Spelling : Use spell-check to make sure your message is clear and easy to understand.
- Be Polite Online : Don't say or do things online that you wouldn't do in person.

HOW TO BE SAFE ONLINE?

The Internet is a great place to have fun and learn, but it's important to stay safe. Here are some safety tips :

- Keep Personal Information Private : Don't share your address, phone number, your parents' work addresses, or your school's name without your parents' permission.
- Keep Your Password Safe : If you join an online game or chat group, never give out your password to anyone.
- Talk to Your Parents : If you see something online that makes you uncomfortable, tell your parents right away.
- Be Careful with Photos and Personal Info : Don't send your picture or personal information to anyone online without asking your parents first.
- Ignore Uncomfortable Messages : Don't reply to messages that make you feel uneasy.

I. Tick (✓) the correct option :

1. The Internet is also called the
   (a) Website ☐ (b) Net ☐
   (c) Web page ☐ (d) Web ☐
2. It is a collection of one or more related web pages linked together through hyperlinks. ☐
   (a) Website ☐ (b) Web server ☐
   (c) Hypertext ☐ (d) ISP ☐
3. It is a program used to locate and display web pages. ☐
   (a) WWW ☐ (b) ARPANET ☐
   (c) Browser ☐ (d) None of these ☐
4. An individual page of a website
   (a) Web page ☐ (b) Web address ☐
   (c) WWW ☐ (d) Search engine ☐
5. The first page of a website
   (a) Web page ☐ (b) Home page ☐
   (c) WWW ☐ (d) None of these ☐

II. Fill in the blanks :

1. The ..... is a huge network of computers around the world.
2. A ..... is a program used to display web pages.
3. The link connecting one web page to another is known as a ..... .
4. The page of a website is called the ..... page.
5. A special set of manners to be followed when using the Internet is called ..... .

III. Write 'T' for true and 'F' for false.

1. You can book rail tickets and shop for things on the Internet. ☐
2. A web page can contain text and icons. ☐

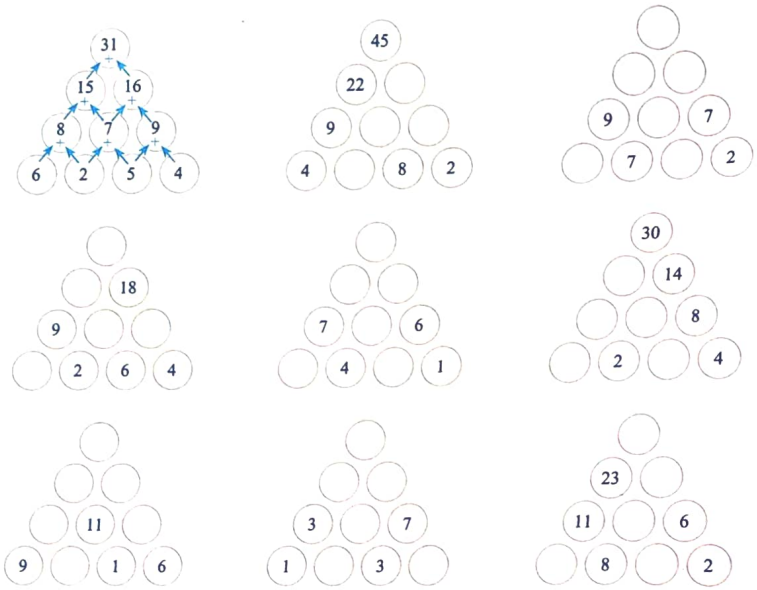
3. When you bring the pointer over hyperlinked image, it changes to a hand symbol.
4. You can give your personal information on the Internet with your parents permission.
5. A hyperlink cannot take you to another website.

IV. Answer the following questions.

1. What is the Internet?
2. What is a browser? Give an example of a browser.
3. Write about a good practice you should follow while using the Internet.
4. Mention any two Internet safety tips you should keep in mind while online.
5. How are web pages related to a website?

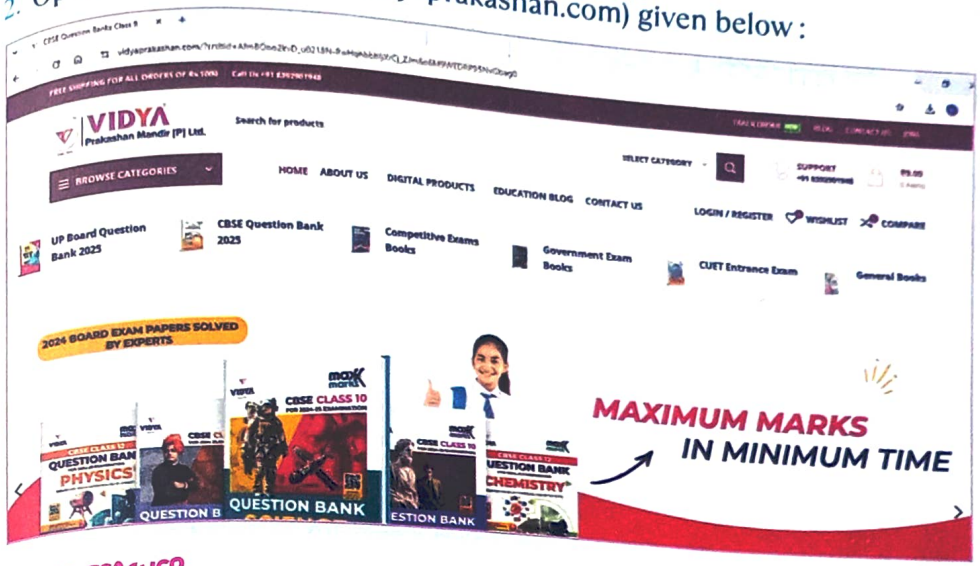
Smart Thinking

❖ **Add two numbers next to each other in each row. Write the answer in the circle above two numbers.**



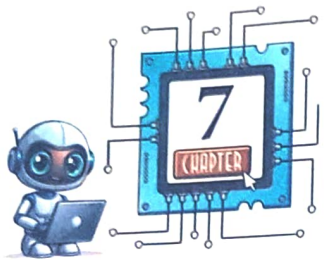
Lab Activity

1. Open the website of your school.
2. Open the website (www.vidyaprakashan.com) given below :



FOR THE TEACHER

Explain the concept of interent and its uses in detail.



Learning Scratch

Let's Recall

Name the following ScratchJr icons :

....

In this Chapter

- Starting Scratch
- Scratch Window
- Adding a Block
- Changing Costumes
- Adding Sound
- Starting a Project
- Making the sprite speak
- Adding Another Character

WHAT IS CODING?

Coding is the process of creating instructions for computers using programming languages. It is the way we communicate with machines, which allows us to create software like mobile apps, general programs, and operating systems. In this chapter, we will learn to code in a program known as Scratch.

WHAT IS SCRATCH?

This year, we're going to learn about Scratch, a beginner-friendly programming language designed to help young students understand programming concepts. Scratch is free to download, and the latest version is 3.29.1. You can get it from this link

https://scratch.mit.edu/download

In Scratch, you work with 'sprites' which are drawings you can control. Sprites can move, change size, play sounds, and more. They can represent anything, like a ball, a butterfly or an animal.

You create programs in Scratch by dragging and connecting blocks that have commands on them. These programs are called scripts. When you click on a script, Scratch runs the blocks from the top to bottom.

Using Scratch, you can create simple animations and games. In this chapter, you will learn to make a sprite move, turn, and add sounds.

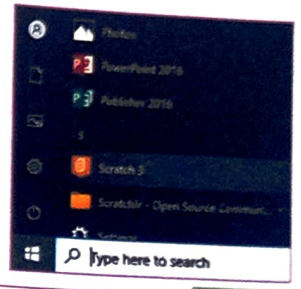
Starting Scratch

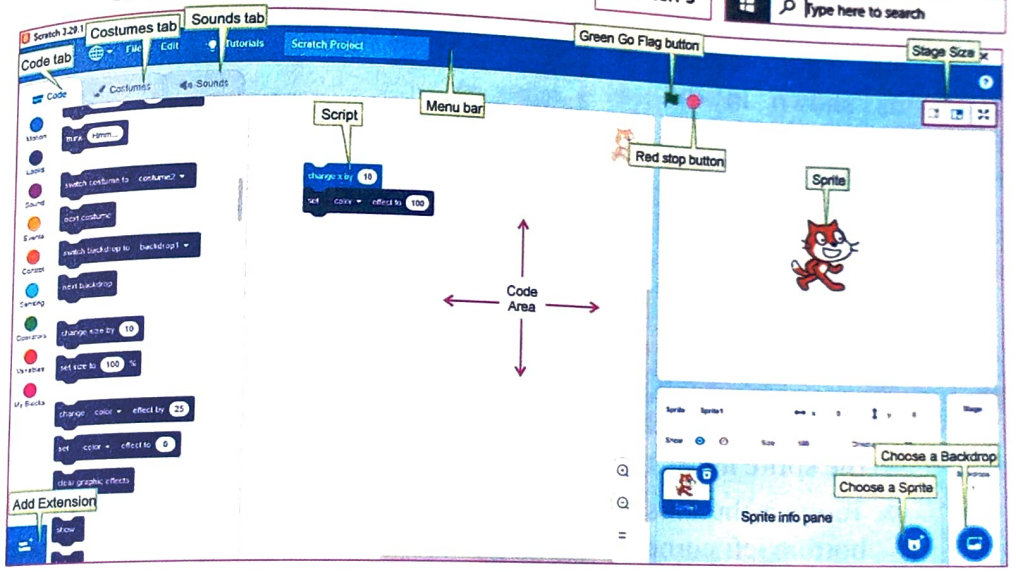
To start Scratch 3, follow these steps :

Step 1. Click on the Start button.

Step 2. Select Scratch 3.

Or Double-click the Scratch 3 icon on the desktop. This will open the Scratch window.





Parts of the Scratch Window

1. Stage : The stage is where everything happens. You can add different backgrounds to set the scene.

2. Sprite : A sprite is a character on the stage that performs actions. You can add as many characters as you want to the stage.
3. Script Area : The script area is where you create code using programming blocks. To program a sprite, you simply click and drag blocks into the Script area.
4. Programming Palette : This is where all the programming blocks are stored. The blocks are organized into sections like Motion, Looks, Sound, etc.

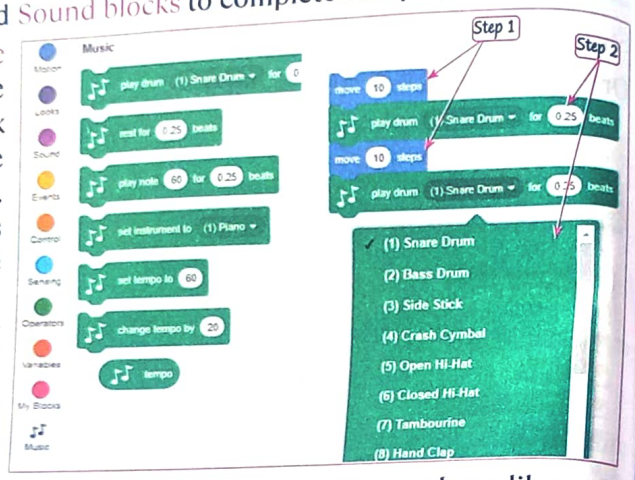
Now that we know the different parts of Scratch, let's start coding!

PROJECT : SPRITE DANCE

Now, let's create a program to make the sprite dance in Scratch. We'll use the Looks, Motion, Control, and Sound blocks to complete this project.

Step 1. Drag the move blocks from the Motion block category into the programming area. Set the values as shown in the example.

Step 2. Drag the play drum blocks from the Music block category. Click on the drop-down arrow on the play drum block and choose a drum sound you like.



Tip :

- The Motion block category is light blue and has blocks that make the sprite move, turn, go to a specific position and more.
- To open the Music block category, click on the button in the bottom left corner of the Scratch window, then select Music from the available options.

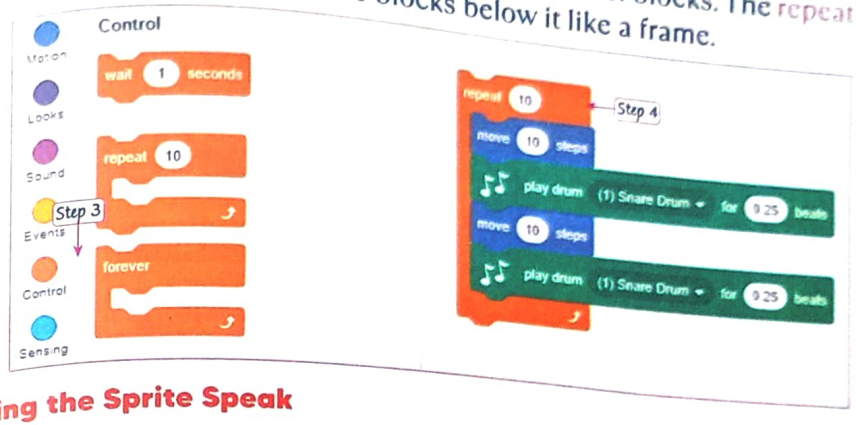
Now, the sprite will move and play the drum sound, but it only plays the drum once.

Wouldn't you like to make it move and play the drum continuously?

Adding a Repeat block : This block helps you repeat your code over and over. You'll find it in the Control block category, which is orange.

Step 3. Click on the Control block category.

Step 4. Drag a repeat block and place it over the other blocks. The repeat block should wrap around the blocks below it like a frame.

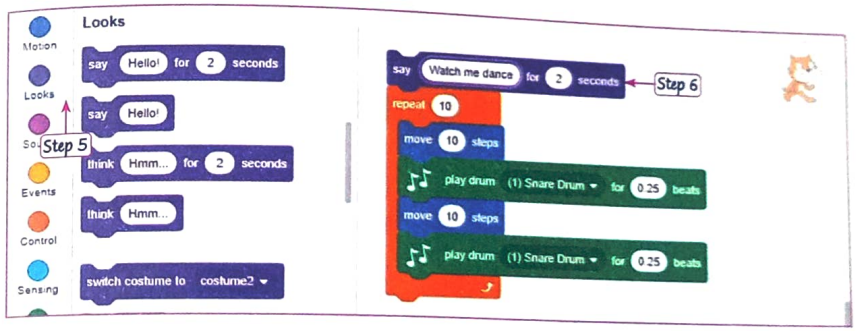


Making the Sprite Speak

To make the sprite say something on the stage, use the say block. This block allows the sprite to display speech bubbles and is found in the Looks block category, which is blue.

Step 5. Click on the Looks block category.

Step 6. Drag out a say block, change the text, and place it at the top of the stack.

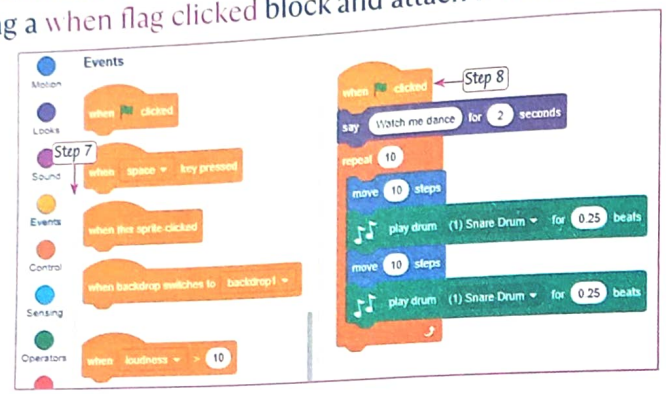


Tip :

- When you click the Green Flag, your script will start running. This block is found in the Events block category and is yellow.
- To stop the script, click the Stop button.

Step 7. Click on the Events scripts.

Step 8. Drag a when flag clicked block and attach it to the top of the stack.

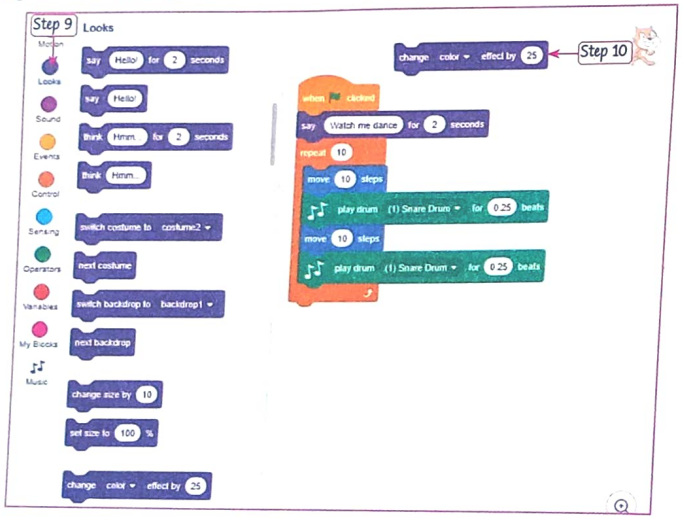


Animating the sprite with colour change

To make the sprite change colours :

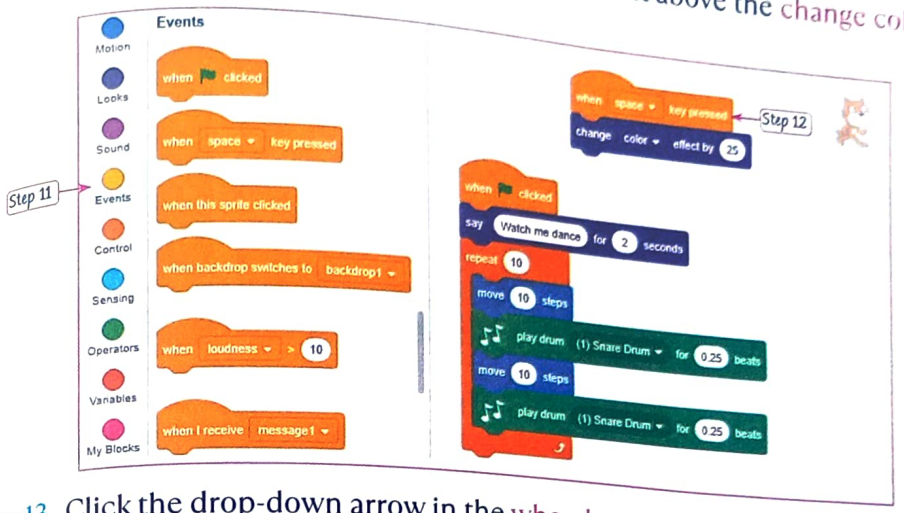
Step 9. Click on the Looks block category.

Step 10. Drag out a change color effect by block into the script area.

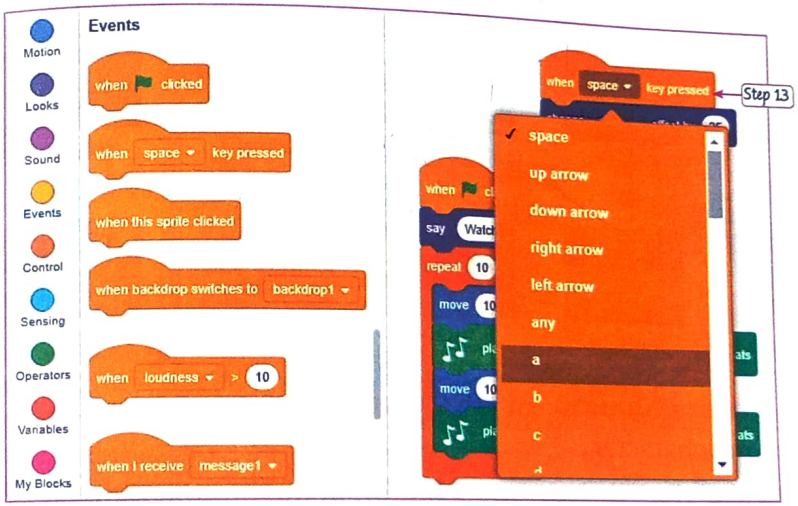


Step 11. To make the cat change color when you press a key on the keyboard, you can use the when key pressed block from the Events block category. Click on the Events block category.

Step 12. Drag the when key pressed block and place it above the change color effect by block.



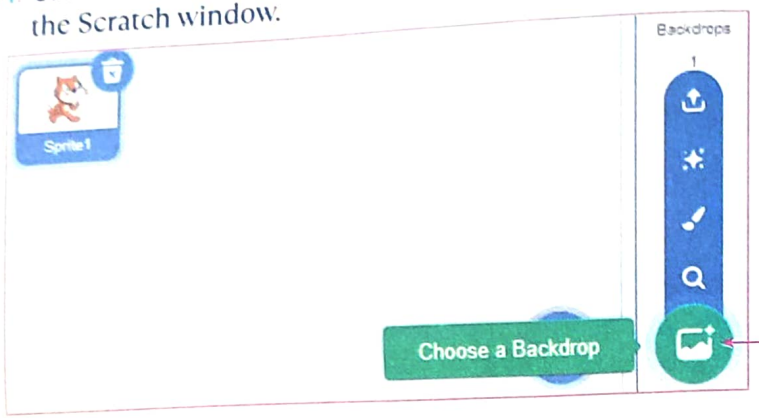
Step 13. Click the drop-down arrow in the when key pressed block and choose any key you like.



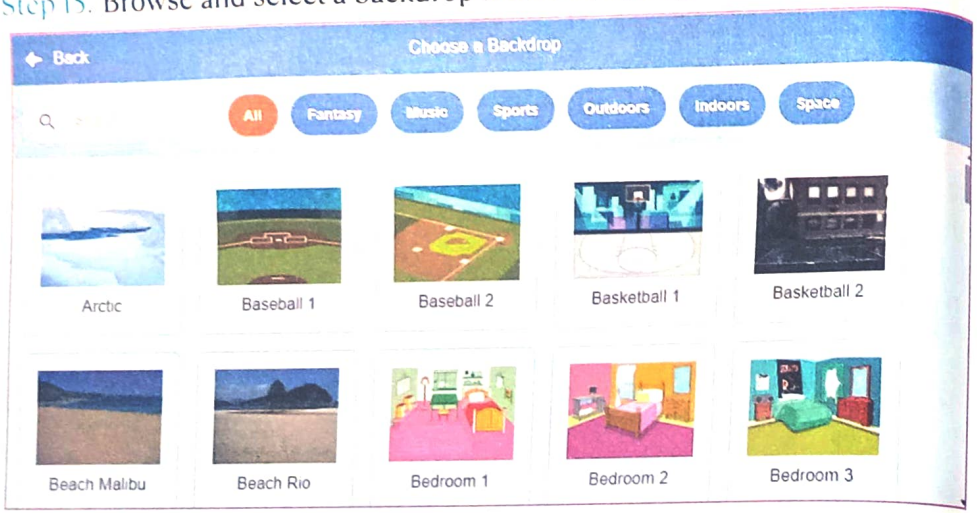
Adding a Background

To make the stage more interesting, you can add a backdrop.

Step 14. Click on the Choose a Backdrop button in the bottom right corner of the Scratch window.



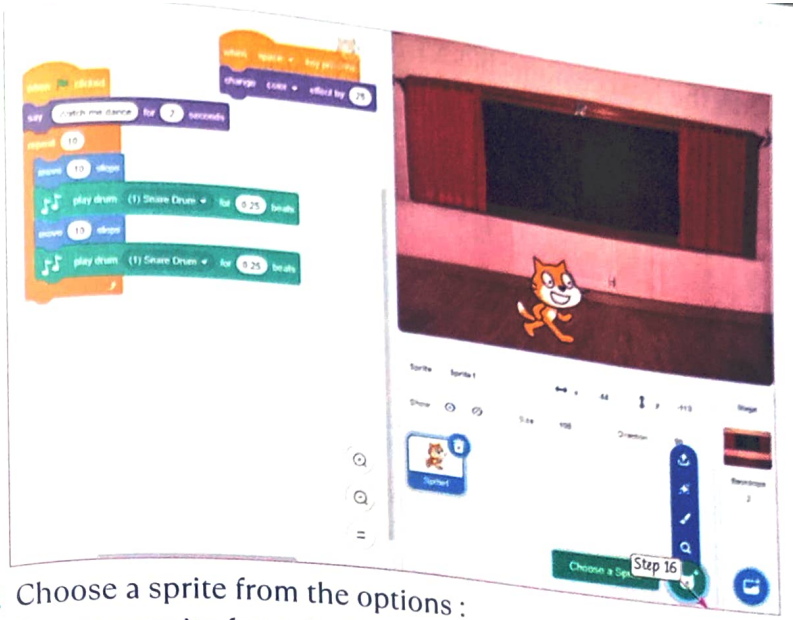
Step 15. Browse and select a backdrop from the Library.



Adding Another Character on the Stage

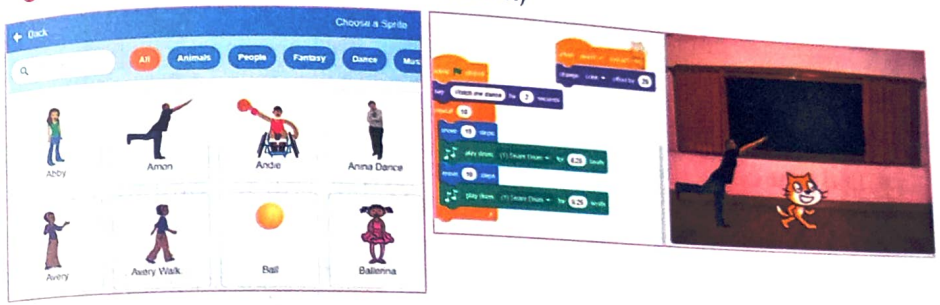
To add another character :

Step 16. Click on the Choose a Sprite button in the bottom right corner of the stage.



Step 17. Choose a sprite from the options :

- Choose a sprite from the Library
- Paint your own sprite
- Upload your own image or sprite
- Surprise (random sprite selection)



Animating the New Sprite

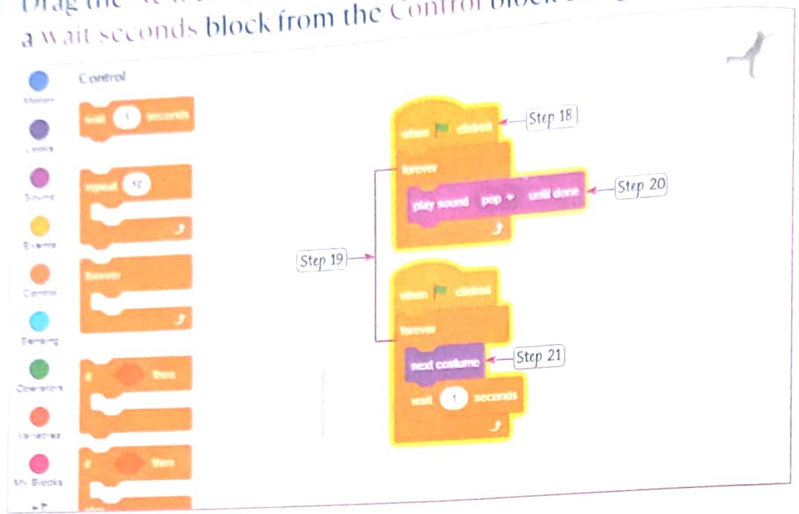
To make the new sprite move and change its appearance :

Step 18. Drag the when flag clicked block from the Events block category.

Step 19. Drag the forever block from the Control block category.

Step 20. Drag the Play sound block from the Sound block category and add it to your script.

Step 21 Drag the Next costume block from the Looks block category. Also, add a wait seconds block from the Control block category.

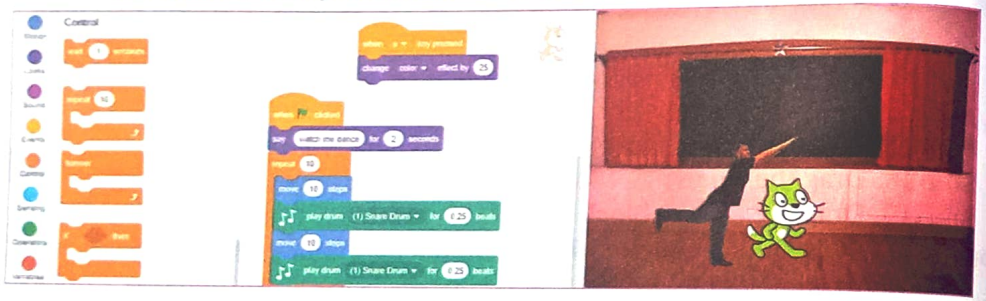


Tip :

- The wait block slows down the costume change, so you can see the animation.

Step 22. Click the Green flag to run your animation and enjoy your sprite's dance!

Did you enjoy the dance? Now it's time to save your project. You can also share it with your friends or the Scratch online community.



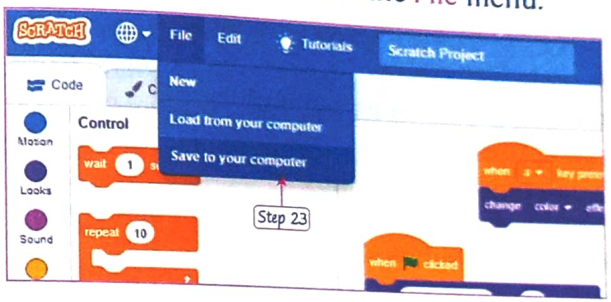
Activity Corner

Crack the code to find the answer to the questions about penguins.

A	B	C	D	E	F	G	H	I	J	K	L	M
1	2	3	4	5	6	7	8	9	10	11	12	13
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
14	15	16	17	18	19	20	21	22	23	24	25	26

1. What do penguins eat?
   $\overline{11}$ $\overline{18}$ $\overline{9}$ $\overline{12}$ $\overline{12}$, $\overline{19}$ $\overline{17}$ $\overline{21}$ $\overline{9}$ $\overline{4}$ and $\overline{6}$ $\overline{9}$ $\overline{19}$ $\overline{8}$
2. What is the smallest species of penguin called?
   $\overline{12}$ $\overline{9}$ $\overline{20}$ $\overline{20}$ $\overline{12}$ $\overline{5}$, $\overline{2}$ $\overline{12}$ $\overline{21}$ $\overline{5}$ and $\overline{16}$ $\overline{5}$ $\overline{14}$ $\overline{7}$ $\overline{21}$ $\overline{9}$ $\overline{14}$
3. How fast can some penguins swim?
   $\overline{6}$ $\overline{9}$ $\overline{6}$ $\overline{20}$ $\overline{5}$ $\overline{5}$ $\overline{14}$ $\overline{13}$ $\overline{16}$ $\overline{8}$
4. How many different species of penguins are there?
   $\overline{14}$ $\overline{9}$ $\overline{14}$ $\overline{5}$ $\overline{20}$, $\overline{5}$ $\overline{5}$ $\overline{14}$

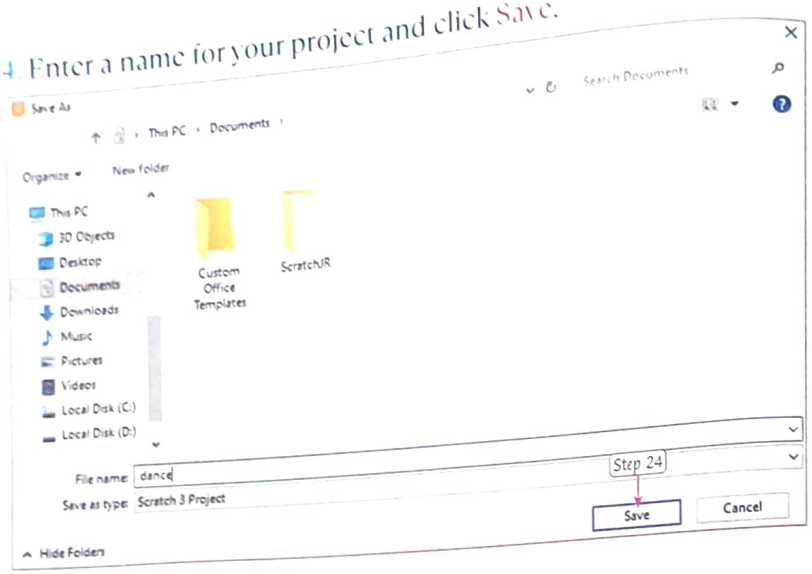
Step 23. Select Save to your computer from the File menu.



Let's Get Better

Anu wants to organise a family trip to a hill station for a durtion of four days. Her family intends to visit five different places during their stay. Help her create plan to ensure they have an enjoyable experience.

Step 24. Enter a name for your project and click Save.



I. Tick (✓) the correct option :

1. What is the primary purpose of coding?
   (a) To decorate web pages
   (b) To create instructions for computers
   (c) To draw sprites
   (d) To design physical machines
2. Which category in Scratch contains the say block, which allows a sprite to speak in speech bubbles?
   (a) Control (b) Motion
   (c) Looks (d) Events
3. Where can you find the when key pressed block in Scratch?
   (a) Looks (b) Control
   (c) Events (d) Sound
4. Which block should you use to repeat actions multiple times in Scratch?
   (a) Forever block (b) Repeat block
   (c) Next costume block (d) When flag clicked block
5. Which step should you follow to save your Scratch project?
   (a) Click the Green Flag
   (b) Select Save to your computer from the File menu
   (c) Drag the next costume block
   (d) Choose a sprite

II. Fill in the blanks :

1. We can make the cat dance 10 times by adding the ..... block.
2. We can add dialogues for the cat by adding the ..... block.
3. Background can be added to stage by ..... button.
4. ..... flag helps to start the program.
5. ..... button stops the program.
6. ..... block helps to execute the program endlessly.
7. ..... button helps to add more Sprites to the stage.

III. Write 'T' for true and 'F' for false :

1. We can make the cat dance 30 times by forever block.
2. We cannot create our own backdrop or sprite.
3. The cat stops dancing if stop button is clicked.
4. Any sprite can change its costume using the next backdrop block.
5. The cat can say "How are you?" using the Repeat block.
6. The when flag clicked block can be added from the Events block category.

IV. Answer the following questions.

1. Write briefly about the following elements of the Scratch window.
   (a) Stage (b) Sprite
   (c) Programming Pallette (d) Script
2. Write the names of four New Sprite buttons.
3. How can we add a new backdrop the stage?

Smart Thinking

- See the image given below. Multiplication is given of some numbers. Colour the blocks in which the unit's digit of the product is 0 or 5.

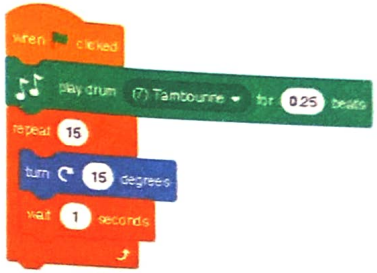
Lab Activity

- Develop a coir of Star Fishes using underwater backdrop. Each Star Fish plays different drum when keys of the keyboard are pressed.

- Develop a project to animate your name.

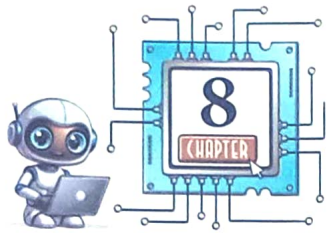
The steps are :

- Choose the first letter.
- Add colour effects and play drum.
- Add a backdrop
- Add the next letter
- Make it spin.
- Go on adding letters and make them spin or change their colour.



FOR THE TEACHER

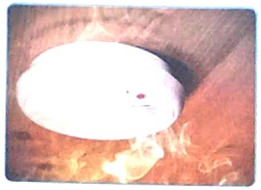
- Help the students to open Scratch 3.29.1 window.
- Explain to the students about drag & drop the blocks in the programming area.
- Help the students to explore the changes that can be done with each of the block.



Artificially Intelligent Gadgets

Let's Recall

Name the following AI enabled devices :



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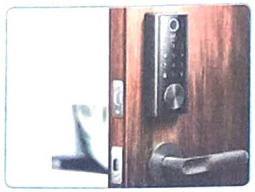

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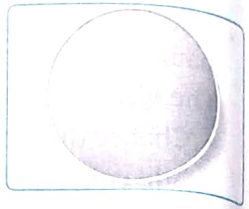

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In this Chapter

- A Brief History of AI and Robots
- Actuators: Making Machines Move
- AI in Gadgets/Devices
- Applications of AI
- Sensors and Their Role
- Robotics

In your previous class, you were introduced to the fascinating world of Artificial Intelligence (AI). You learned how AI mimics human behaviour by learning, thinking, and making decisions. AI is everywhere, from our homes to our daily routines, helping us manage tasks and improve our lives. In this lesson, we'll explore AI in gadgets and devices, and how it is becoming an integral part of our everyday lives.

HISTORY OF AI AND ROBOTS

The idea of machines that can think and act like humans has been around for a long time. The concept of artificial intelligence first gained attention in the 1950s when computer scientists began exploring how machines could

be programmed to perform tasks that required human intelligence, such as playing chess or solving problems.

One of the earliest AI programs was created by Alan Turing, a famous mathematician and computer scientist. He proposed the idea of a machine that could simulate any human task, and this idea eventually led to the development of AI.

Robots, on the other hand, have a longer history. The word 'robot' was first used in a play called R.U.R. (Rossum's Universal Robots) by Czech writer Karel Čapek in 1920. The play imagined a world where robots, made to serve humans, eventually rebel against their creators. While robots in reality haven't rebelled, they have become incredibly useful in industries, homes, and even space exploration.

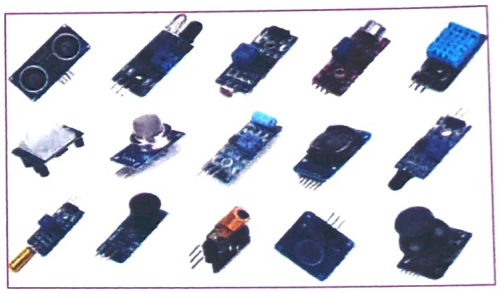
As technology advanced, AI and robotics combined to create intelligent machines that could learn from their environment and improve over time. Today, AI-powered robots can perform complex tasks and are used in many areas, including healthcare, manufacturing, and entertainment.

AI IN GADGETS/DEVICES

We encounter numerous AI-powered gadgets every day, making our lives more convenient and efficient. These devices can understand their surroundings and respond accordingly. Have you ever wondered how a robot vacuum knows where to clean, or how a smart thermostat adjusts the temperature automatically? The secret lies in the sensors and actuators that these gadgets use.

WHAT IS A SENSOR?

A sensor is a device that detects changes in the environment and provides the information needed for a machine to make decisions. Just like our senses (like touch, sight, and hearing), sensors allow AI gadgets to sense what is happening around them. Different types of sensors are used in different



Types of sensors

gadgets, depending on the task they need to perform. Here are some common types of sensors :

Position Sensors : Detect the location or position of an object.

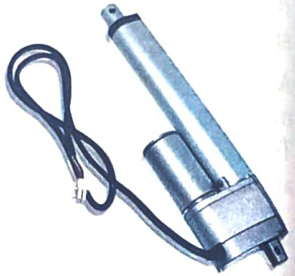
Pressure Sensors : Measure the force exerted on a surface.

Temperature Sensors : Sense the temperature in a room or object.

Vibration Sensors : Detect vibrations or movements.

These sensors work together to provide data to the gadget, which then processes this information to perform specific actions.

WHAT IS AN ACTUATOR?



An actuator is the part of a machine that makes things move. Once the sensors provide the necessary data, the actuator responds by performing physical actions. For example, an actuator in a robot might rotate its arms or move its legs. In simpler terms, actuators are the muscles of a smart device, allowing it to interact with the physical world.

For instance, in robotic arms used in manufacturing, actuators control the movement of the arm to assemble parts. In smart homes, actuators in devices like smart door locks or smart curtains open and close them as needed.

APPLICATIONS OF AI

AI has several applications today. Some of them are discussed below :

Smart Homes

In this new era of artificial intelligence, a smart home is a house filled with smart devices. These devices are connected to the internet and can be controlled from your smartphone.

Examples include smart door locks, doorbells, TVs, thermostats, lights, refrigerators, and washing machines.

Smart homes are much more convenient and intelligent than regular homes. They save energy, provide better safety and security, and can be customised

to fit your lifestyle. The main goal of a smart home is to make your life more comfortable.

Advantages of Smart Homes

Here are some benefits of having a smart home :

- Control Everything from One Place : Manage all your smart devices from a single app on your phone.
- Add New Devices Easily : You can easily add new smart devices and appliances whenever you want.
- Save Energy : Smart homes are designed to be energy-efficient, which helps save power.
- Enhanced Safety and Security : They offer better security and protection for your home.
- Remote Control : You can monitor and control your devices from anywhere, even when you're not at home.

Let's Get Better

Smart homes are different from normal homes. Suppose, you have to redesign your home into a smart home then list the devices that you need to replace.

Smart Security

Smart Security takes care of all your safety needs automatically. It uses sensors and smart devices like cameras, fire and smoke detectors, automated alarms, and smart gates to protect your home or school. For example, imagine a school gate that only opens for authorised people like students or teachers. Smart Security is more reliable than human guards as it avoids mistakes.

Advantages of Smart Security

Here are some benefits of smart security :

- No Human Errors : It eliminates mistakes that can happen with human security guards.
- Immediate Alerts : It sends warnings and alerts to your phone so you can act quickly.
- Lower Costs : It reduces the need for human security, saving money.
- Better Protection : It offers more effective security to keep your place safe.

Smart Cars

Smart cars are driverless vehicles that can drive themselves from one place to another. They use sensors to navigate traffic safely, avoid collisions, and follow traffic rules. Smart cars can also detect traffic signals and signs. These cars are already being used in developed countries like the US, with Tesla being one of the well-known manufacturers.

Advantages of Smart Cars

Here are some benefits of smart cars :

- No Driving Mistakes : They avoid errors that human drivers might make.
- Automatic Control : They handle all driving controls by themselves.
- Better Fuel Efficiency : They use fuel more efficiently than cars driven by humans.
- Follow Traffic Rules : They always obey traffic rules and never break them.

Augmented Reality

Augmented Reality (AR) is a technology that mixes digital information with the real world. It uses special apps and programs to overlay virtual objects or information onto our physical surroundings. AR apps on smartphones use GPS to find your location, and advanced AR systems used in military training can recognise objects and gestures.

AR changes how we see the world by adding virtual elements to it.

Applications of Augmented Reality :

- Gaming : AR makes games more exciting by bringing game elements into the real world.
- Medical Procedures : AR helps make surgeries easier and reduces the risk of complications.
- Eye Health : The Eye Decide app uses AR to help patients learn about their eye health.

Virtual Reality

Virtual Reality (VR) is a new and exciting technology that creates an artificial environment using software. It tricks your mind into thinking that this digital world is real. The term virtual reality means almost real. Just like we experience the real world through our senses, VR provides a computer-generated environment that we can interact with.

VR creates the feeling that you are in a different place. It mainly uses sight and sound to make objects in the virtual world seem real. The simplest VR experience is a 3D image on a computer that you can explore using the keyboard or mouse to move or zoom in on the image.

Applications of Virtual Reality

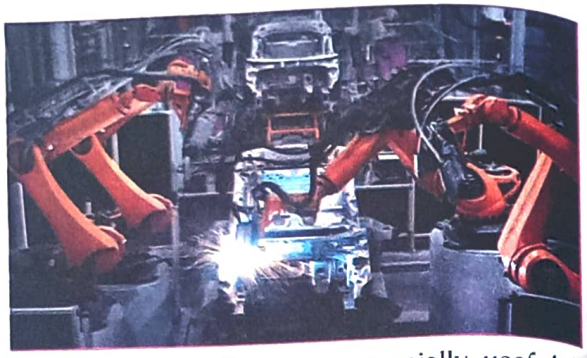
- Architecture : Helps in designing and visualising buildings.
- Sports : Provides training simulations for athletes.
- Medicine : Assists in medical training and treatment planning.
- Arts : Creates immersive art experiences.
- Entertainment : Offers engaging video games and virtual tours.

In summary, VR creates a virtual environment that feels real to our senses. It has a wide range of uses for both fun and serious purposes, and we can expect even more exciting applications in the future.

Robotics

Robotics is a field of engineering that combines mechanical engineering, electronics, computer science, and more. It involves designing, building, and

controlling robots, which are machines that can perform tasks automatically.



Robots are becoming increasingly important and might replace some human activities in the future. They can take on many forms, but many are designed to look and act like humans, walking, talking, and responding like us. Robots are especially useful in dangerous situations, such as bomb detection and defusing. Some robots are inspired by nature and are called nature-inspired robots.

In 1940, writer Isaac Asimov introduced the term 'robotics' in a story and created three guiding principles for robots, known as the Laws of Robotics :

- Robots will never harm humans.
- Robots will protect themselves but not break the rules.
- Robots will follow human commands.

Advantages of Robots

Robots have many benefits :

- They work automatically : Robots operate without needing human help.
- They are precise : Robots complete tasks without making mistakes.
- They gather difficult information : Robots can access information that's hard for humans to get.
- They respond quickly : Robots act faster than people.
- They handle extreme conditions : Robots can work in harsh weather conditions like heat, cold, or rain.

Disadvantages of Robots

However, robots also have some downsides :

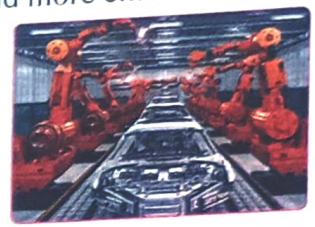
- They need power : Robots require electricity to function.
- Job loss : People might lose jobs if robots replace them in factories.

- High maintenance : Robots need regular maintenance to avoid malfunctioning.
- Expensive upkeep : Maintaining robots can be costly.

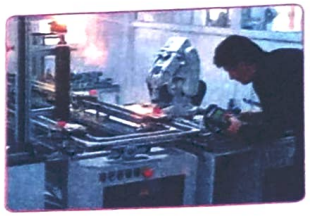
Applications of Robots

Robots are used in many areas to make tasks easier and more efficient:

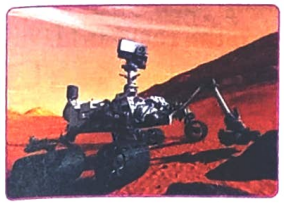
In Industry : Robots are used for tasks like welding, painting, assembling, and packaging. They can work in extreme temperatures and handle delicate tasks on circuit boards.



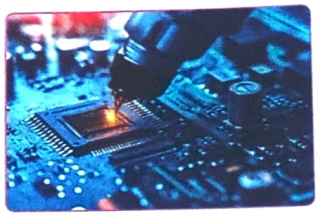
In Medicine : Robots can help doctors examine patients remotely, assist in surgeries from afar, and transport medicines and meals in hospitals.



In Scientific Research : Robots are used to find new treatments, create artificial organs, and analyse cells in labs.



In Space : Space robots can move around, take measurements, collect samples, and send back photos from space.



In Manufacturing ICs : Robots help make integrated circuits, placing tiny electronic parts accurately and efficiently.

I. Tick (✓) the correct option:

1. What does AI in gadgets often use to understand their surroundings?
   (a) Actuators
   (b) Sensors
   (c) Virtual Reality
   (d) Augmented Reality
2. What is the main purpose of a smart home?
   (a) To improve the aesthetics of a house
   (b) To make home life more convenient and efficient
   (c) To replace all human activities
   (d) To create a manual control system
3. Which of the following is NOT an advantage of smart cars?
   (a) They make driving mistakes.
   (b) They are more fuel efficient.
   (c) They follow traffic rules.
   (d) They handle driving controls automatically.
4. What does Augmented Reality (AR) primarily add to the real world?
   (a) Physical objects
   (b) Digital information
   (c) Virtual reality environments
   (d) Human interactions
5. What term did Isaac Asimov introduce related to robotics?
   (a) Automation
   (b) Artificial Intelligence
   (c) Robotics
   (d) Cybernetics

II. Fill in the blanks :

1. A ..... is a device that detects changes in the environment and provides information for a machine to make decisions.
2. Smart homes use devices that can be controlled remotely through a ..... .
3. An ..... is the part of a machine that makes things move based on data from sensors.
4. The term 'virtual reality' means '..... real'.
5. Isaac Asimov's Laws of Robotics include the rule that robots must never harm ..... .

III. Give Two Examples of Each

1. Give two examples of AI-powered gadgets.
2. Give two applications of augmented reality.
3. Give two examples of industries where robots are used.
4. Give two types of sensors used in smart devices.
5. Give two benefits of using smart homes.

IV. Short Answer Questions

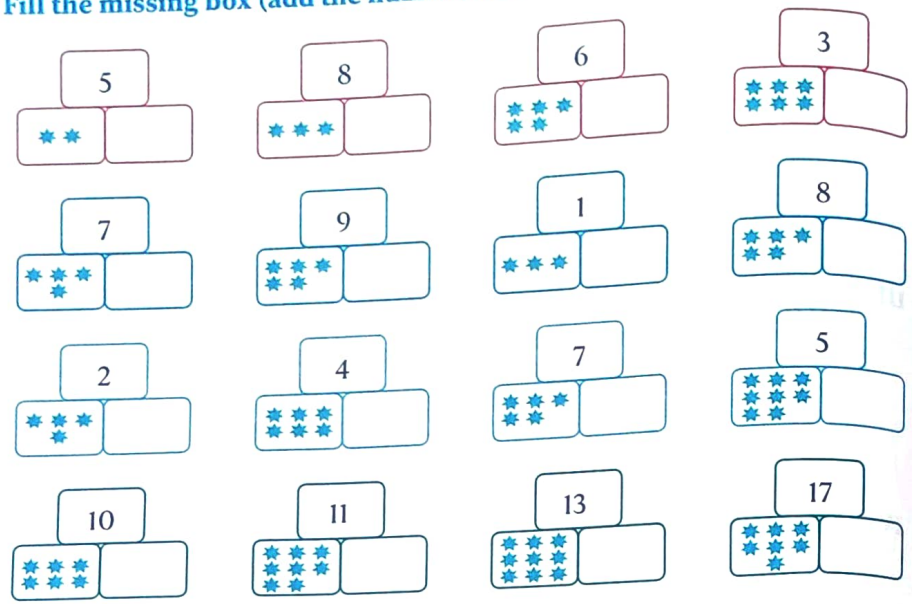
1. What is the role of sensors in AI-powered gadgets?
2. Name one advantage of using smart security systems.
3. What does a smart car use to navigate and avoid collisions?
4. How does Augmented Reality (AR) enhance gaming experiences?
5. What is a common application of virtual reality in the field of medicine?

V. Long Answer Questions

1. Explain how artificial intelligence and robotics have evolved over time. Include key milestones and their impact on modern technology.
2. Describe the role and importance of sensors and actuators in AI-powered gadgets. How do they work together to perform tasks?
3. Discuss the advantages and disadvantages of smart cars. How do these vehicles impact driving safety and efficiency?
4. Explain the concept of Virtual Reality (VR) and its applications. How does VR create immersive experiences and what are some examples of its use?

Smart Thinking

❖ **Fill the missing box (add the number and number of stars).**



FOR THE TEACHER

Help students understand the concepts of AI and robotics by providing real-life examples of how AI assists us in our daily life.

Periodic Test Term-1

(Based on Chapters 1 and 2)

Time _____

Marks _____

Note : All questions are compulsory.

I. Tick (✓) the correct option:

1. Which of the following components of the desktop is illustrated alongside?
   (a) Dustbin ☐ (b) Basket ☐
   (c) Recycle Bin ☐ (d) Bucket ☐
2. Which of the following is not an operating system?
   (a) Windows 10 ☐ (b) Windows 8 ☐
   (c) Windows 11 ☐ (d) Windows 9 ☐
3. The long horizontal bar available at the bottom of the screen is called ..... .
   (a) Taskbar ☐ (b) Sidebar ☐
   (c) Bottom bar ☐ (d) Title bar ☐

II. Write 'T' for true and 'F' for false:

1. The deleted files, folders or icons are stored in documents.
2. You can see the time displayed on the left hand side of the taskbar.
3. The control panel is used to change settings and add or remove programs.

III. Fill in the blanks:

1. ..... is the world's oldest computing device.
2. The ..... is known as the first mechanical and automatic calculator.
3. Concept of modern computers are based on the design of ..... .

IV. Answer the following questions :

1. Name the machine invented by John Napier and write a few words about it.
2. Name two machines which were developed by Charles Babbage?
3. Write a short note on Operating System.

Half Yearly Test Paper

(Based on Chapters 1 to 4)

Time .....

Marks .....

Note : *All questions are compulsory.*

I. Tick (✓) the correct option:

1. Which of these commands places the selected text slightly above the baseline?
   (a) Subscript ☐ (b) Indent
   (c) Superscript ☐ (d) None of these ☐
2. Which of these is the white space all around the printed area of the paper?
   (a) Page orientation ☐ (b) Page size
   (c) Page margin ☐ (d) All of these ☐
3. Which of the following displays the slides as thumbnails?
   (a) Outline pane ☐ (b) Slides pane
   (c) work area ☐ (d) Ribbon Shortcut Section ☐
4. In which tab, Save As option is present?
   (a) Insert ☐ (b) File
   (c) Format ☐ (d) View ☐
5. Which of the following is the shortcut keys to insert a new slide?
   (a) Crtl + M ☐ (b) Alt + M
   (c) Shift + M ☐ (d) Shift + N ☐

II. Write 'T' for true and 'F' for false :

1. Paragraph spacing cannot be changed in a document.
2. Header is placed at the bottom of a page.
3. Shadow effect makes the text more attractive and looks like 3-D text.
4. You cannot change the font type of the text in slides.
5. The content of the presentation is inserted in text placeholder.

III. Fill in the blanks with the help of given box :

1. The ..... option places the selected text slightly below the baseline.
2. The ..... is places at the end of a document.
3. The first window that opens up in PowerPoint is called ..... view.
4. ..... is located at the top of the PowerPoint window.
5. ..... button is used to exit the PowerPoint.

IV. Answer the following questions :

1. Define footer.
2. What is PowerPoint?
3. How do you use the Find and Replace feature of Word?
4. What is a slide? Write steps to create Title slide.
5. What is a placeholder?

V. Complete the given steps to apply shadow to the text.

1. Select the ..... .
2. Click on ..... .
3. Click on ..... command in the ..... group.
4. Select option ..... from the drop-down menu.
5. Click on any ..... .

VI. Fill the boxes given below as instructed :

Number	10 More	10 Less	100 More	100 Less
133				
467				
733				
947				
569				
799				
667				
113				

Periodic Test Term-2

(Based on Chapters 5 and 6)

Time .....

Marks .....

Note : *All questions are compulsory.*

I. Tick (✓) the correct option:

1. Which of these is dotted-line containers on slide layout?
   (a) Placeholders ☐ (b) Animation ☐
   (c) Clipart ☐ (d) None of these ☐
2. The Internet is also called the
   (a) Website ☐ (b) Net ☐
   (c) Web page ☐ (d) Web ☐
3. Which of the following allows you to create text effects that is not possible through formating?
   (a) WordArt ☐ (b) Picture ☐
   (c) Animation ☐ (d) Transitions ☐
4. It is a program used to locate and display web pages.
   (a) WWW ☐ (b) ARPANET ☐
   (c) Browser ☐ (d) None of these ☐

II. Fill in the blanks:

1. ..... view is used to deliver your presentation to the audience.
2. A ..... is a program used to display web pages.
3. PowerPoint 2016 includes ..... in-built style layouts.

III. Write 'T' for true and 'F' for false:

1. There are six different ways to view your presentation. ☐
2. You can book rail tickets and shop for things on the Internet. ☐
3. Slide show view displays miniatures of your slides. ☐

IV. Answer the following questions :

1. What is slide layout? How many slide layouts are available in PowerPoint?
2. What is the Internet?
3. Write the difference between slide sorter view and reading view.

Annual Test Paper

Time .....

Marks .....

Note : *All questions are compulsory.*

I. Tick (✓) the correct option:

1. What is the primary purpose of coding?
   (a) To decorate web pages ☐
   (b) To create instructions for computers ☐
   (c) To draw sprites ☐
   (d) To design physical machines ☐
2. What does AI in gadgets often use to understand their surroundings?
   (a) Actuators ☐
   (b) Sensors ☐
   (c) Virtual Reality ☐
   (d) Augmented Reality ☐
3. Which category in Scratch contains the say block, which allows a sprite to speak in speech bubbles?
   (a) Control ☐ (b) Motion ☐
   (c) Looks ☐ (d) Events ☐

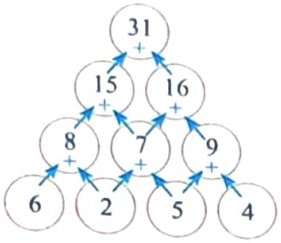
II. Fill in the blanks :

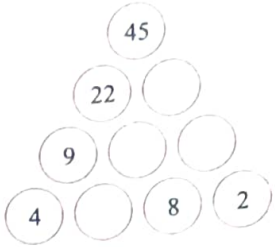
1. We can make the cat dance 10 times by adding the ..... block.
2. A ..... is a device that detects changes in the environment and provides information for a machine to make decisions.
3. We can add dialogues for the cat by adding the ..... block.

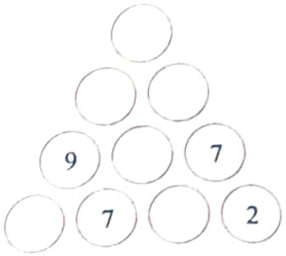
III. Match the following :

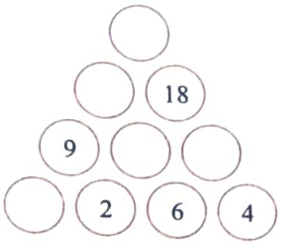
1.	(a) Reading view
2.	(b) Slide Show
3.	(c) Normal view
4.	(d) Slide Sorter view
5.	(e) Outline view button

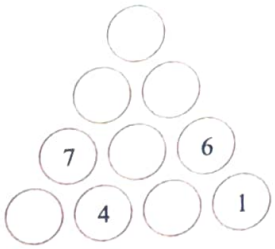
IV. Add two numbers next to each other in each row. Write the answer in the circle above two numbers.

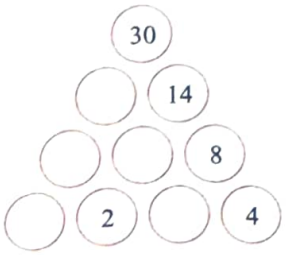


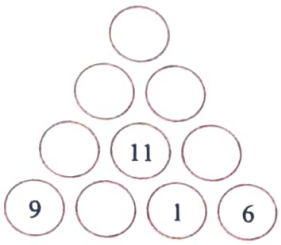


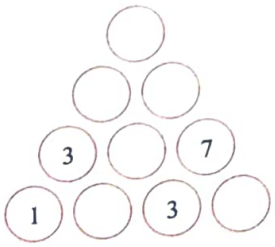


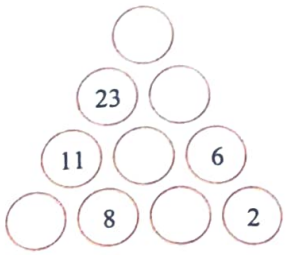












V. Write 'T' for true and 'F' for false :

1. We can make the cat dance 30 times by forever block.
2. One of the earliest AI programs was created by Alan Turing.
3. You cannot change the font of the WordArt.
4. A web page can contain text and icons.

VI. Answer the following questions :

1. Write briefly about the following elements of the Scratch window.
   (a) Stage (b) Sprite
   (c) Programming Pallette (d) Script
2. Explain how artificial intelligence and robotics have evolved over time. Include key milestones and their impact on modern technology.
3. What is the role of sensors in AI-powered gadgets?

Mastering CODING

AI AND ROBOTICS

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